

This IC is a 2-wire serial E<sup>2</sup>PROM with temperature sensor for DIMM serial presence detect which operates in 2.2 V to 3.6 V voltage ranges. This IC has the capacity of 4 K-bit and the organization of 2 pages × 256-word × 8-bit. Page write and sequential read are available.

This IC operates with the I<sup>2</sup>C-bus at 1.0 MHz maximum.

A substantial reduction in current consumption may be achieved by using the software programmed shutdown mode which can be set by the I<sup>2</sup>C-bus.

**Caution** This product is intended to use in general electronic devices such as consumer electronics, office equipment, and communications devices. Before using the product in medical equipment or automobile equipment including car audio, keyless entry and engine control unit, it is imperative to contact our sales representatives.

## ■ Features

### E<sup>2</sup>PROM block

- Page write: 16 bytes / page
- Sequential read
- Write protect function during low power supply voltage
- Write protect: Individual software data protection for each of four 128-byte blocks
- Endurance: 10<sup>6</sup> cycle / word\*1 (Ta = +25°C)
- Data retention: 100 years (Ta = +25°C)
- Memory capacity: 4 K-bit
- Initial delivery state: FFh

### Temperature sensor block

- Temperature accuracy: 0.5°C typ. (Ta = +75°C to +95°C)  
1.0°C typ. (Ta = +40°C to +125°C)
- Temperature sample rate: 8 samples / s min.
- Selectable hysteresis width: No hysteresis, 1.5°C, 3.0°C, 6.0°C

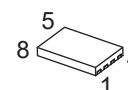
### Overall

- JEDEC standard compliant: TSE2004B2
- Current consumption:
  - E<sup>2</sup>PROM in standby mode and temperature sensor in shutdown mode: 3.0 μA max.
  - E<sup>2</sup>PROM in standby mode and temperature sensor in active mode: 0.1 mA max.
  - E<sup>2</sup>PROM in read operation mode and temperature sensor in active mode: 0.4 mA max.
  - E<sup>2</sup>PROM in write operation mode and temperature sensor in active mode: 2.0 mA max.
- Operation voltage range: 2.2 V to 3.6 V
- Operation frequency: 1.0 MHz max. (V<sub>DD</sub> = 2.2 V to 3.6 V)
- Noise suppression: Schmitt trigger and noise filter on input pins (SCL, SDA)
- Operation temperature range: Ta = -20°C to +125°C
- Lead-free (Sn 100%), halogen-free

\*1. For each address (Word: 8-bit)

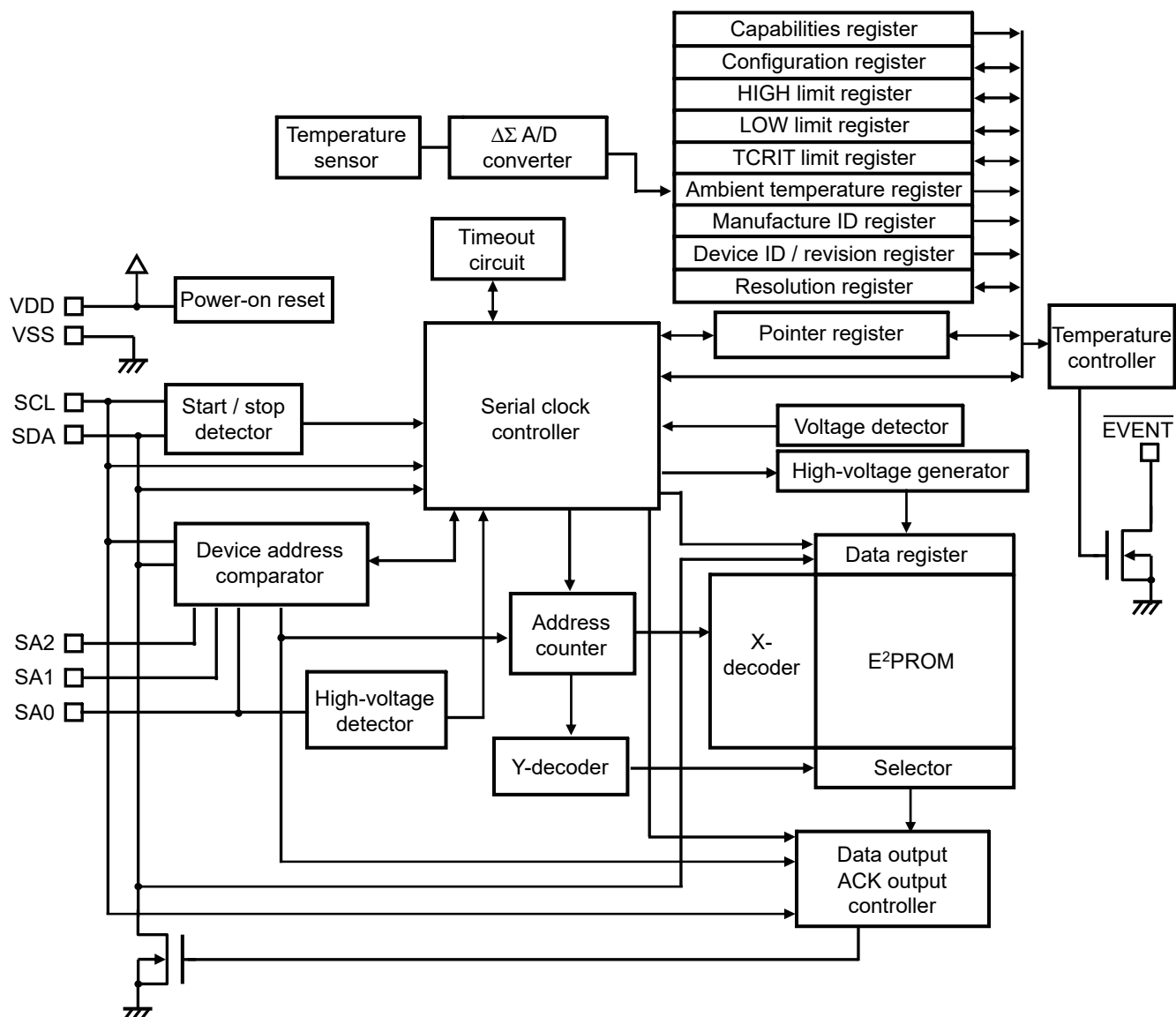
## ■ Package

- DFN-8(2030)B



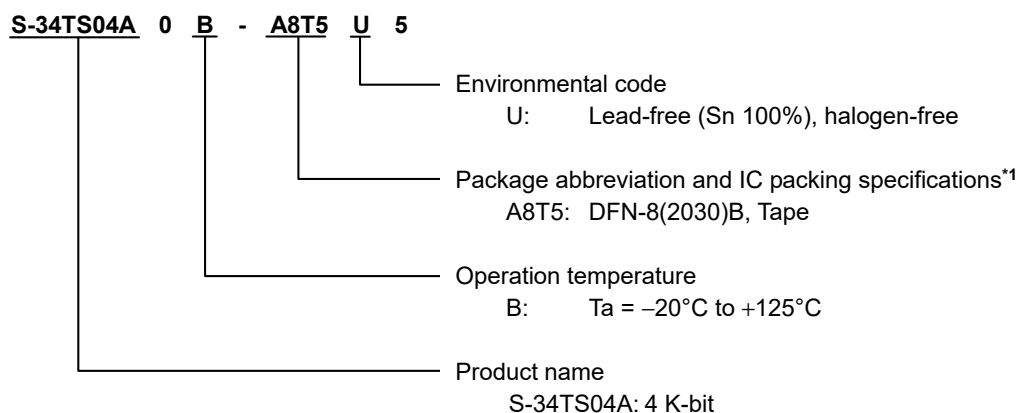
(3.0 × 2.0 × t0.8 mm)

## ■ Block Diagram



## ■ Product Name Structure

### 1. Product name



\*1. Refer to the tape drawing.

### 2. Package

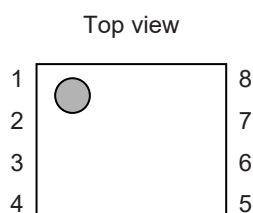
| Package Name | Dimension    | Tape         | Reel         | Land         |
|--------------|--------------|--------------|--------------|--------------|
| DFN-8(2030)B | PQ008-B-P-SD | PQ008-B-C-SD | PQ008-B-R-SD | PQ008-B-L-SD |

### 3. Product name list

| Product Name       | Capacity | Package Name |
|--------------------|----------|--------------|
| S-34TS04A0B-A8T5U5 | 4 K-bit  | DFN-8(2030)B |

## ■ Pin Configuration

### 1. DFN-8(2030)B



| Pin No. | Symbol | Description              |
|---------|--------|--------------------------|
| 1       | SA0    | Select address input     |
| 2       | SA1    | Select address input     |
| 3       | SA2    | Select address input     |
| 4       | VSS    | Ground                   |
| 5       | SDA*1  | Serial data I/O          |
| 6       | SCL*1  | Serial clock input       |
| 7       | EVENT  | Temperature event output |
| 8       | VDD    | Power supply             |

\*1. Do not use it in "High-Z".

**Remark** For DFN-8(2030)B package, connect the heatsink of back side to the board, and set electric potential open or V<sub>SS</sub>. However, do not use it as the function of electrode.

## ■ Absolute Maximum Ratings

Table 1

| Item                             | Symbol           | Absolute Maximum Rating | Unit |
|----------------------------------|------------------|-------------------------|------|
| Power supply voltage             | V <sub>DD</sub>  | −0.3 to +4.3            | V    |
| Input voltage (SCL, SA1, SA2)    | V <sub>IN</sub>  | −0.3 to +4.3            | V    |
| SA0 pin high level input voltage | V <sub>HV</sub>  | −0.3 to +10.0           | V    |
| I/O voltage (SDA)                | V <sub>IO</sub>  | −0.3 to +4.3            | V    |
| Output voltage (EVENT)           | V <sub>OUT</sub> | −0.3 to +4.3            | V    |
| Operation ambient temperature    | T <sub>opr</sub> | −20 to +125             | °C   |
| Storage temperature              | T <sub>stg</sub> | −65 to +150             | °C   |

**Caution** The absolute maximum ratings are rated values exceeding which the product could suffer physical damage. These values must therefore not be exceeded under any conditions.

## ■ Recommended Operation Conditions

Table 2

| Item                             | Symbol           | Condition                                 | Min.                  | Max.                  | Unit |
|----------------------------------|------------------|-------------------------------------------|-----------------------|-----------------------|------|
| Power supply voltage             | V <sub>DD</sub>  | TSE2004B2 compliant                       | 2.2                   | 3.6                   | V    |
| Operation ambient temperature    | T <sub>opr</sub> | TSE2004B2 compliant                       | −20                   | +125                  | °C   |
| High level input voltage         | V <sub>IH</sub>  | −                                         | 0.7 × V <sub>DD</sub> | V <sub>DD</sub> + 0.5 | V    |
| Low level input voltage          | V <sub>IL</sub>  | −                                         | −0.3                  | 0.3 × V <sub>DD</sub> | V    |
| SA0 pin high level input voltage | V <sub>HV</sub>  | V <sub>HV</sub> − V <sub>DD</sub> ≥ 4.8 V | 7.0                   | 10.0                  | V    |

## ■ Pin Capacitance

Table 3

(Ta = +25°C, f = 1.0 MHz, V<sub>DD</sub> = 2.5 V)

| Item               | Symbol           | Condition                                  | Min. | Max. | Unit |
|--------------------|------------------|--------------------------------------------|------|------|------|
| Input capacitance  | C <sub>IN</sub>  | V <sub>IN</sub> = 0 V (SCL, SA0, SA1, SA2) | −    | 6    | pF   |
| I/O capacitance    | C <sub>I/O</sub> | V <sub>I/O</sub> = 0 V (SDA)               | −    | 8    | pF   |
| Output capacitance | C <sub>OUT</sub> | V <sub>OUT</sub> = 0 V (EVENT)             | −    | 8    | pF   |

## ■ Endurance

Table 4

| Item      | Symbol         | Operation Ambient Temperature | Min.            | Max. | Unit           |
|-----------|----------------|-------------------------------|-----------------|------|----------------|
| Endurance | N <sub>w</sub> | Ta = +25°C                    | 10 <sup>6</sup> | −    | cycle / word*1 |

\*1. For each address (Word: 8-bit)

## ■ Data Retention

Table 5

| Item           | Symbol | Operation Ambient Temperature | Min. | Max. | Unit |
|----------------|--------|-------------------------------|------|------|------|
| Data retention | −      | Ta = +25°C                    | 100  | −    | year |

## ■ DC Electrical Characteristics

Table 6

| Item                             | Symbol | Condition                                                                         | Ta = −20°C to +125°C |      | Unit |
|----------------------------------|--------|-----------------------------------------------------------------------------------|----------------------|------|------|
|                                  |        |                                                                                   | VDD = 2.2 V to 3.6 V |      |      |
|                                  |        |                                                                                   | Min.                 | Max. |      |
| Shutdown current consumption     | ISD    | E <sup>2</sup> PROM: standby mode<br>Temperature sensor: shutdown mode            | —                    | 3.0  | μA   |
| Standby current consumption      | IDD1   | E <sup>2</sup> PROM: standby mode<br>Temperature sensor: active mode              | —                    | 0.1  | mA   |
| Current consumption (READ)       | IDDR   | E <sup>2</sup> PROM: read mode<br>Temperature sensor: active mode<br>f = 1.0 MHz  | —                    | 0.4  | mA   |
| Current consumption (WRITE)      | IDDW   | E <sup>2</sup> PROM: write mode<br>Temperature sensor: active mode<br>f = 1.0 MHz | —                    | 2.0  | mA   |
| Input leakage current            | ILI    | SCL, SDA<br>VIN = VSS to VDD                                                      | —                    | 1.0  | μA   |
| Output leakage current           | ILO    | SDA, $\overline{\text{EVENT}}$<br>VOUT = VSS to VDD                               | —                    | 1.0  | μA   |
| Input current 1                  | IIL    | SA0, SA1, SA2<br>VIN < 0.3 × VDD                                                  | —                    | 50.0 | μA   |
| Input current 2                  | IIH    | SA0, SA1, SA2<br>VIN > 0.7 × VDD                                                  | —                    | 2.0  | μA   |
| Input impedance 1                | ZIL    | SA0, SA1, SA2<br>VIN = 0.3 × VDD                                                  | 30                   | —    | kΩ   |
| Input impedance 2                | ZIH    | SA0, SA1, SA2<br>VIN = 0.7 × VDD                                                  | 800                  | —    | kΩ   |
| Low level output voltage         | VOL    | SDA, $\overline{\text{EVENT}}$<br>IOL = 3.0 mA                                    | —                    | 0.4  | V    |
| Low level output current 1       | IOL1   | SDA, $\overline{\text{EVENT}}$<br>VOL = 0.4 V                                     | 20                   | —    | mA   |
| Low level output current 2       | IOL2   | SDA, $\overline{\text{EVENT}}$<br>VOL = 0.6 V, f ≤ 400 kHz                        | 6                    | —    | mA   |
| Power-on reset threshold voltage | VPON   | —                                                                                 | 1.6                  | —    | V    |
| Power-off threshold voltage      | VPOFF  | —                                                                                 | —                    | 0.9  | V    |

## ■ AC Electrical Characteristics

Table 7 Measurement Conditions

|                                   |                                            |
|-----------------------------------|--------------------------------------------|
| Input pulse voltage               | $0.2 \times V_{DD}$ to $0.8 \times V_{DD}$ |
| Input pulse rising / falling time | 20 ns or less                              |
| Output reference voltage          | $0.3 \times V_{DD}$ to $0.7 \times V_{DD}$ |
| Output load                       | 100 pF                                     |

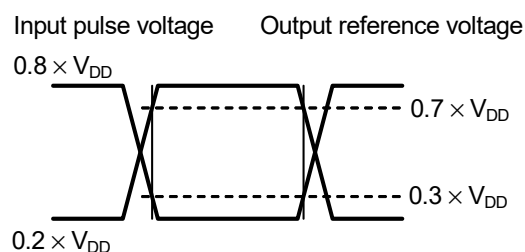


Figure 1 Input / Output Waveform during AC Measurement

Table 8

| Item                       | Symbol   | Ta = −20°C to +125°C |      |          |      | Unit |
|----------------------------|----------|----------------------|------|----------|------|------|
|                            |          | VDD = 2.2 V to 3.6 V |      |          |      |      |
|                            |          | 400 kHz              |      | 1000 kHz |      |      |
|                            |          | Min.                 | Max. | Min.     | Max. |      |
| SCL clock frequency        | fSCL     | 10                   | 400  | 10       | 1000 | kHz  |
| SCL clock time "L"         | tLOW     | 1.3                  | –    | 0.5      | –    | μs   |
| SCL clock time "H"         | tHIGH    | 0.6                  | –    | 0.26     | –    | μs   |
| SCL clock "L" timeout      | tTIMEOUT | 25                   | 35   | 25       | 35   | ms   |
| SCL, SDA rising time       | tR       | 0.02                 | 0.3  | –        | 0.12 | μs   |
| SCL, SDA falling time      | tF       | 0.02                 | 0.3  | –        | 0.12 | μs   |
| Data input setup time      | tSU.DAT  | 100                  | –    | 50       | –    | ns   |
| Data input hold time       | tHD.DI   | 0                    | –    | 0        | –    | ns   |
| Data output hold time      | tHD.DAT  | 200                  | 900  | 0        | 350  | ns   |
| Start condition setup time | tSU.STA  | 0.6                  | –    | 0.26     | –    | μs   |
| Start condition hold time  | tHD.STA  | 0.6                  | –    | 0.26     | –    | μs   |
| Stop condition setup time  | tSU.STO  | 0.6                  | –    | 0.26     | –    | μs   |
| Bus release time           | tBUF     | 1.3                  | –    | 0.5      | –    | μs   |
| Noise suppression time     | ti       | –                    | 50   | –        | 50   | ns   |
| Power-off time             | tPOFF    | 1                    | –    | 1        | –    | ms   |
| Initialize time            | tINIT    | 0.2                  | –    | 0.2      | –    | ms   |

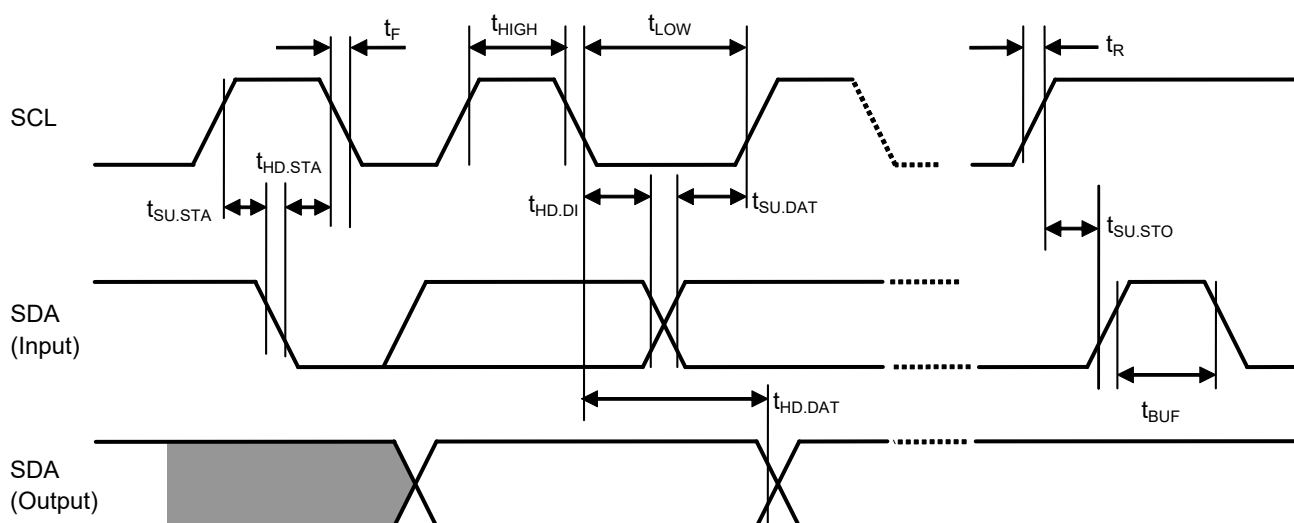


Figure 2 Bus Timing

Table 9

| Item       | Symbol | Ta = −20°C to +125°C |      | Unit |
|------------|--------|----------------------|------|------|
|            |        | VDD = 2.2 V to 3.6 V |      |      |
|            |        | Min.                 | Max. |      |
| Write time | tWR    | —                    | 5.0  | ms   |

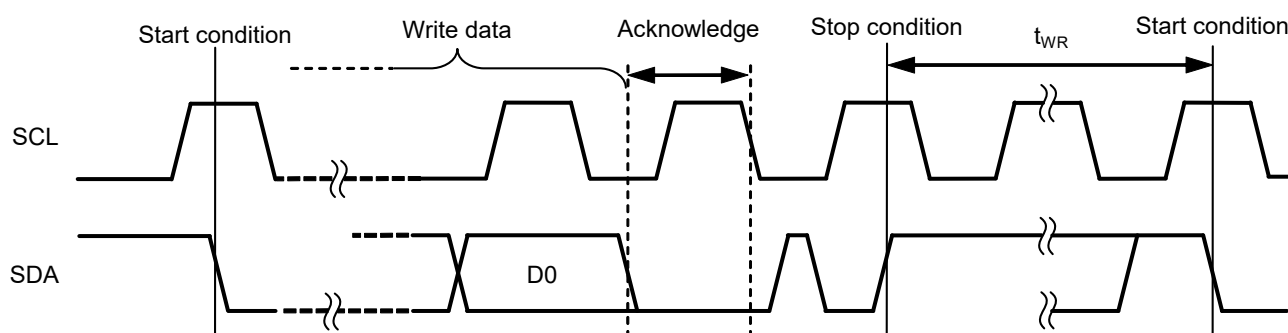


Figure 3 Write Cycle Timing

## ■ Temperature Characteristics

Table 10

| Item                        | Symbol            | Condition               | V <sub>DD</sub> = 2.2 V to 3.6 V |      |      | Unit |
|-----------------------------|-------------------|-------------------------|----------------------------------|------|------|------|
|                             |                   |                         | Min.                             | Typ. | Max. |      |
| Temperature sensor accuracy | T <sub>ACC1</sub> | Ta = +75°C to +95°C     | —                                | ±0.5 | ±1.0 | °C   |
|                             | T <sub>ACC2</sub> | Ta = +40°C to +125°C    | —                                | ±1.0 | ±2.0 | °C   |
|                             | T <sub>ACC3</sub> | Ta = -20°C to +125°C    | —                                | ±2.0 | ±3.0 | °C   |
| Resolution                  | T <sub>RES</sub>  | Default value           | —                                | 0.25 | —    | °C   |
| Temperature conversion time | t <sub>CONV</sub> | All TRES [1:0] settings | —                                | —    | 125  | ms   |

## ■ Pin Functions

### 1. VDD (Power supply) pin

The VDD pin is used to apply positive supply voltage. Regarding the applied voltage value, refer to "■ Recommended Operation Conditions". Set a bypass capacitor of about 0.1  $\mu$ F between the VDD pin and the VSS pin for stabilization as close to IC as possible.

### 2. SA0, SA1 and SA2 (Select address input) pins

In this IC, to set the slave address, connect each of the SA0 pin, SA1 pin and SA2 pin to the VSS pin or the VDD pin. Therefore the users can set 8 types of slave address by a combination of the SA0 pin, SA1 pin, SA2 pin.

Comparing the slave address transmitted from the master device and one that you set, makes possible to select one slave address from other devices connected onto the bus.

Each of the SA0 pin, SA1 pin and SA2 pin has a built-in pull-down resistor. In open, the pin is set to the same status as it connected to the VSS pin.

The SA0 pin is used to detect the V<sub>HV</sub> voltage, when decoding an SWPn or CWP instruction. Refer to **Table 11** for pin setting and device select code.

### 3. SDA (Serial data I/O) pin

The SDA pin is used for the bi-directional transmission of serial data. This pin is a signal input pin, and an Nch open-drain output pin.

In use, generally, connect the SDA line to any other device which has the open-drain or open-collector output with Wired-OR connection by pulling up to V<sub>DD</sub> by a resistor.

### 4. SCL (Serial clock input) pin

The SCL pin is used for the serial clock input. Since the signals are processed at a rising or falling edge of the SCL clock, pay attention to the rising and falling time and comply with the specification.

### 5. $\overline{\text{EVENT}}$ (Temperature event output) pin

The  $\overline{\text{EVENT}}$  pin is an open-drain output that requires a pull-up resistor to V<sub>DD</sub> on the system motherboard or integrated into the master controller.  $\overline{\text{EVENT}}$  pin has three operating modes, depending on the configuration register settings, and the output status is defined by the measured temperature and the temperature limit register setting. These modes are interrupt, comparator, or TCRIT only.

**Figure 4** shows an example of the measured temperature versus time, with the corresponding behavior of the  $\overline{\text{EVENT}}$  pin in each of these three modes.

#### 5.1 Interrupt mode

In interrupt mode, the  $\overline{\text{EVENT}}$  pin will be asserted when the measured temperature crosses any temperature limit setting. Once the pin has been asserted, the pin will remain asserted until it is released by writing "1" to the CLEAR bit in the configuration register. After the pin is de-asserted, the pin will be re-asserted when the measured temperature crosses any temperature limit setting.

#### 5.2 Comparator mode

In comparator mode, the  $\overline{\text{EVENT}}$  pin will be asserted when the measured temperature is above the high limit or below "the low limit – hysteresis width". The pin will clear itself when the measured temperature is below "the high limit – hysteresis width" and above the low limit.

#### 5.3 TCRIT only mode

In TCRIT only mode, the  $\overline{\text{EVENT}}$  pin will only be asserted if the measured temperature exceeds the critical temperature. Once the pin has been asserted, it will remain asserted until the temperature drops below "the critical temperature – hysteresis width".

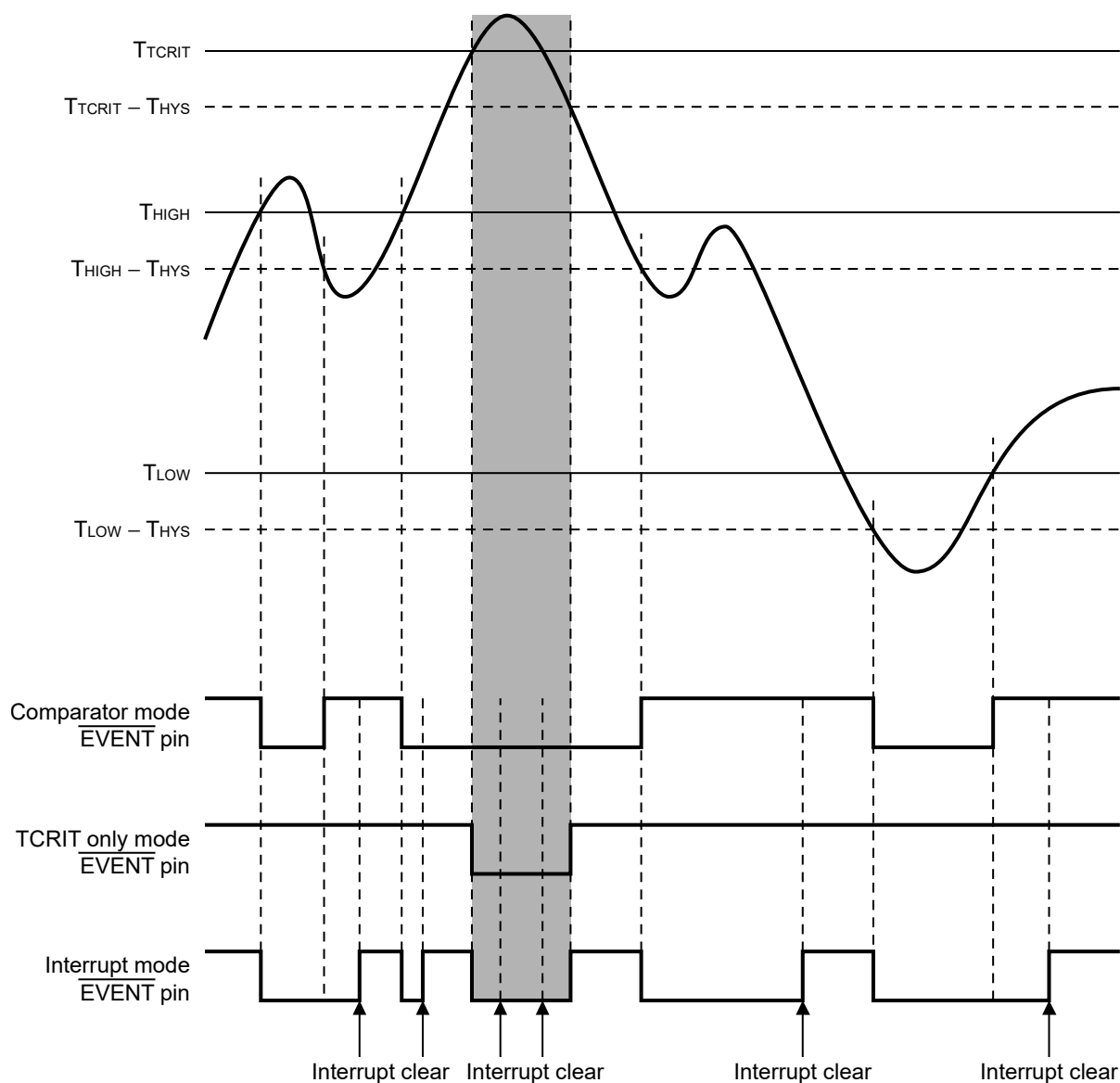


Figure 4  $\overline{\text{EVENT}}$  pin (Active "L")

## ■ Initial Delivery State

Initial delivery state of all address E<sup>2</sup>PROM is "FFh". All write protects are cleared.

## ■ Operation

This IC behaves as a slave device in the I<sup>2</sup>C-bus protocol.

This IC has unrelated two functions as 4 K-bit E<sup>2</sup>PROM with software write protection and temperature sensor measuring ambient temperature.

All operations are synchronized by the serial clock. Read and write operations are initiated by a start condition, generated by the master device. The start condition is followed by a device select code and read / write bit, and this IC generates an acknowledge bit.

The 7-bit device select code is constructed of 4-bit device type identifier code (DTIC) and 3-bit code which shows the state of the SA0 pin, SA1 pin, and SA2 pin. DTIC is a code to define functions.

When writing data to this IC, this IC generates an acknowledge bit during the 9th bit time, following the master device's 8-bit transmission. When data is read by the master device, the master device acknowledges the receipt of the data byte in the same way. Data transfers are terminated by a master device which generates stop condition after an acknowledge for write, and after no acknowledge for read.

This IC has the timeout function. This IC shall not initiate clock stretching, which is an optional I<sup>2</sup>C-bus feature.

### 1. Setting of DTIC

Operation function is identified by DTIC. There are three types of DTIC of which two are used for E<sup>2</sup>PROM and one is used for temperature sensor. The E<sup>2</sup>PROM memory may be accessed using a DTIC of "1010b", and to perform the software write protection or the E<sup>2</sup>PROM page address operations a DTIC of "0110b" is required. The temperature sensor registers are accessed using a DTIC of "0011b".

### 2. E<sup>2</sup>PROM function

This IC has a 4 K-bit E<sup>2</sup>PROM array. The E<sup>2</sup>PROM array is divided into two pages consisting of a lower 256-byte page and an upper 256-byte page, which change over by the set E<sup>2</sup>PROM page address order. Each page has two 128-byte blocks. Each block can be set to write-protected by software write protect function. Page write operation up to 16 bytes and sequential read operation are available.

All operations of the E<sup>2</sup>PROM section are inhibited to be performed during E<sup>2</sup>PROM write time.

For more detail, refer to "■ E<sup>2</sup>PROM Operation".

### 3. Temperature sensor function

This IC has a digital temperature sensor. With the nine types of temperature sensor registers, this IC can be performed to read measured ambient temperature from register and to write configuration data to register. This IC can output comparison result of ambient temperature and specified limit value from  $\overline{\text{EVENT}}$  pin.

Temperature sensor regularly refresh temperature register value after power-on, and can go into shutdown condition when SHDN bit is set. During shutdown condition, temperature sensor circuit is stopped and holds temperature sensor register value, reducing current consumption.

All operations of the temperature sensor section are performed anytime.

For more detail, refer to "■ Temperature Sensor Operation".

#### 4. Initialization operation after power-on

By a power-on reset circuit, this IC initializes the internal circuit at the time of power-on. Perform the beginning (start condition) of the instruction transmission to this IC after the initialization by the power-on reset circuit. Regarding the details of power-on reset, refer to "■ Reset and Initialization".

#### 5. Start condition

Start is identified by a "H" to "L" transition of the SDA line while the SCL line is stable at "H". Every operation begins from a start condition.

#### 6. Stop condition

Stop is identified by a "L" to "H" transition of the SDA line while the SCL line is stable at "H". When a device receives a stop condition during a read sequence, the read operation is interrupted, and the device enters standby mode. When a device receives a stop condition during an E<sup>2</sup>PROM write sequence, the reception of the write data is halted, and this IC initiates an E<sup>2</sup>PROM write cycle.

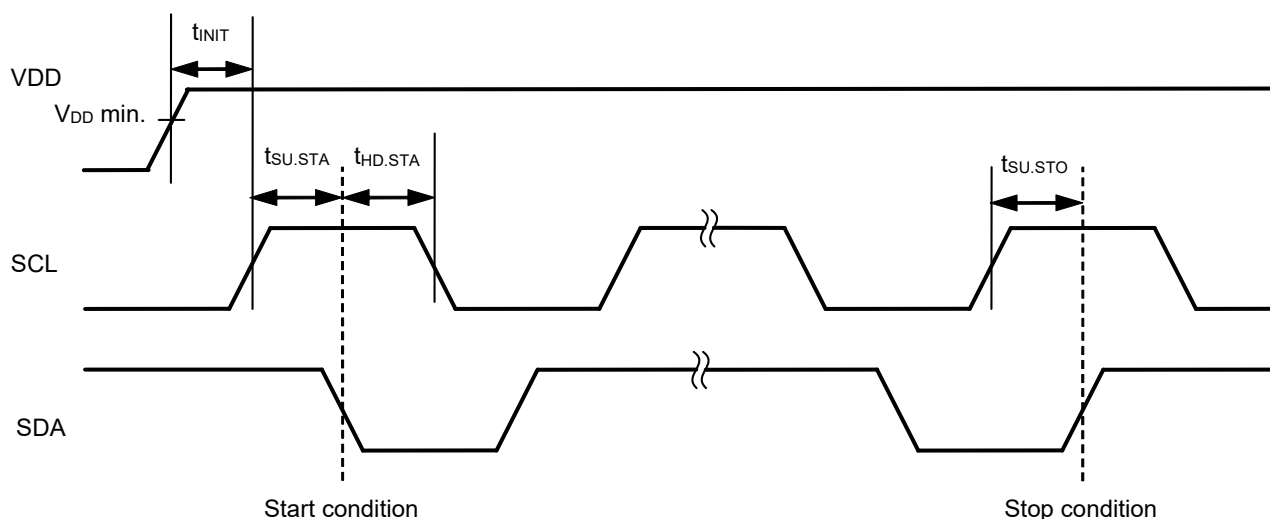
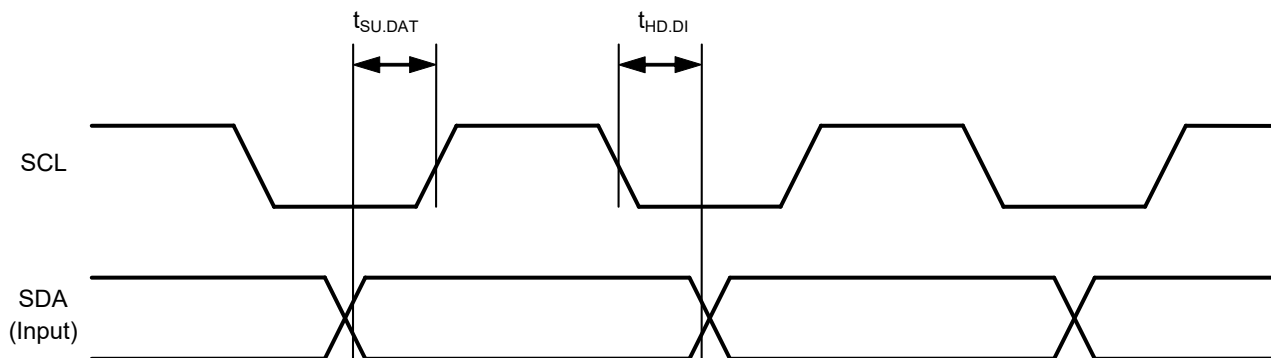


Figure 5 Start / Stop Conditions after Power-on

## 7. Data transmission

Changing the SDA line while the SCL line is "L", data is transmitted.

Changing the SDA line while the SCL line is "H", a start or stop condition is recognized.

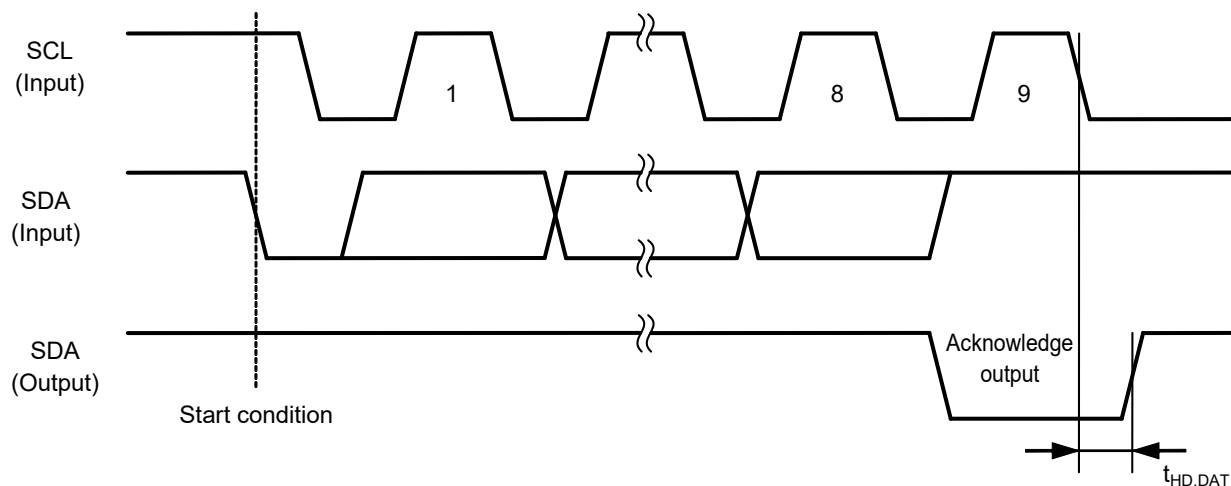


**Figure 6 Data Transmission Timing**

## 8. Acknowledge

The unit of data transmission is 8 bits. During the 9th clock cycle period the receiver on the bus pulls down the SDA line to acknowledge the receipt of the 8-bit data.

When an E<sup>2</sup>PROM write cycle is in progress, this IC does not generate an acknowledge for DTIC of "1010b" and "0110b".



**Figure 7 Acknowledge Output Timing**

## 9. Device addressing

To start communication, the master device on the system generates a start condition to the slave device. Following this, the master device sends the device select code.

Table 11 Device Select Code

| Instruction                                 | Device Type Identifier |    |    |    | Select Address Signal |     |     | R/W | SA Pin |     |                 |
|---------------------------------------------|------------------------|----|----|----|-----------------------|-----|-----|-----|--------|-----|-----------------|
|                                             | B7                     | B6 | B5 | B4 | B3                    | B2  | B1  | B0  | SA2    | SA1 | SA0             |
| Read / write E <sup>2</sup> PROM*1          | 1                      | 0  | 1  | 0  | SA2                   | SA1 | SA0 | R/W | SA2    | SA1 | SA0             |
| Set write protection, block 0 (SWP0)        | 0                      | 1  | 1  | 0  | 0                     | 0   | 1   | 0   | —*2    | —*2 | V <sub>HV</sub> |
| Set write protection, block 1 (SWP1)        | 0                      | 1  | 1  | 0  | 1                     | 0   | 0   | 0   | —*2    | —*2 | V <sub>HV</sub> |
| Set write protection, block 2 (SWP2)        | 0                      | 1  | 1  | 0  | 1                     | 0   | 1   | 0   | —*2    | —*2 | V <sub>HV</sub> |
| Set write protection, block 3 (SWP3)        | 0                      | 1  | 1  | 0  | 0                     | 0   | 0   | 0   | —*2    | —*2 | V <sub>HV</sub> |
| Clear write protection for all blocks (CWP) | 0                      | 1  | 1  | 0  | 0                     | 1   | 1   | 0   | —*2    | —*2 | V <sub>HV</sub> |
| Read SWP0 status (RPS0)                     | 0                      | 1  | 1  | 0  | 0                     | 0   | 1   | 1   | —*2    | —*2 | —*2             |
| Read SWP1 status (RPS1)                     | 0                      | 1  | 1  | 0  | 1                     | 0   | 0   | 1   | —*2    | —*2 | —*2             |
| Read SWP2 status (RPS2)                     | 0                      | 1  | 1  | 0  | 1                     | 0   | 1   | 1   | —*2    | —*2 | —*2             |
| Read SWP3 status (RPS3)                     | 0                      | 1  | 1  | 0  | 0                     | 0   | 0   | 1   | —*2    | —*2 | —*2             |
| Set page address to 0 (SPA0)                | 0                      | 1  | 1  | 0  | 1                     | 1   | 0   | 0   | —*2    | —*2 | —*2             |
| Set page address to 1 (SPA1)                | 0                      | 1  | 1  | 0  | 1                     | 1   | 1   | 0   | —*2    | —*2 | —*2             |
| Read page address (RPA)                     | 0                      | 1  | 1  | 0  | 1                     | 1   | 0   | 1   | —*2    | —*2 | —*2             |
| Read / write temperature register*1         | 0                      | 0  | 1  | 1  | SA2                   | SA1 | SA0 | R/W | SA2    | SA1 | SA0             |

\*1. Slave addresses (SA2, SA1, SA0) are compared by the select address input pins (SA2, SA1, SA0) of a memory device with the address value which is set beforehand.

\*2. Connected to the VSS pin or VDD pin.

The device select code consists of a 4-bit device type identifier, and 3-bit select signals (SA2 pin, SA1 pin, SA0 pin). To address the memory array, the 4-bit device type identifier is "1010b"; to access the write-protection settings and the page address settings, it is "0110b"; and to access the temperature sensor settings is "0011b".

Up to eight devices can be connected on a single I<sup>2</sup>C-bus. Address select signals (SA2 pin, SA1 pin, SA0 pin) should be set on each device, respectively. The device continues the operation only when the received device select code matches the address select signal.

The SWPn, CWP, RPSn write protect commands and SPAn, RPA E<sup>2</sup>PROM page address commands do not use the select address, therefore all devices on the I<sup>2</sup>C-bus will act on these commands simultaneously.

The 8th bit is the read / write bit (R/W). This bit is set to "1" for read and "0" for write operations. If a match occurs on the device select code, the corresponding device gives an acknowledge on serial data (SDA) during the 9th bit time. If the device does not match the device select code, this IC goes into standby mode automatically.

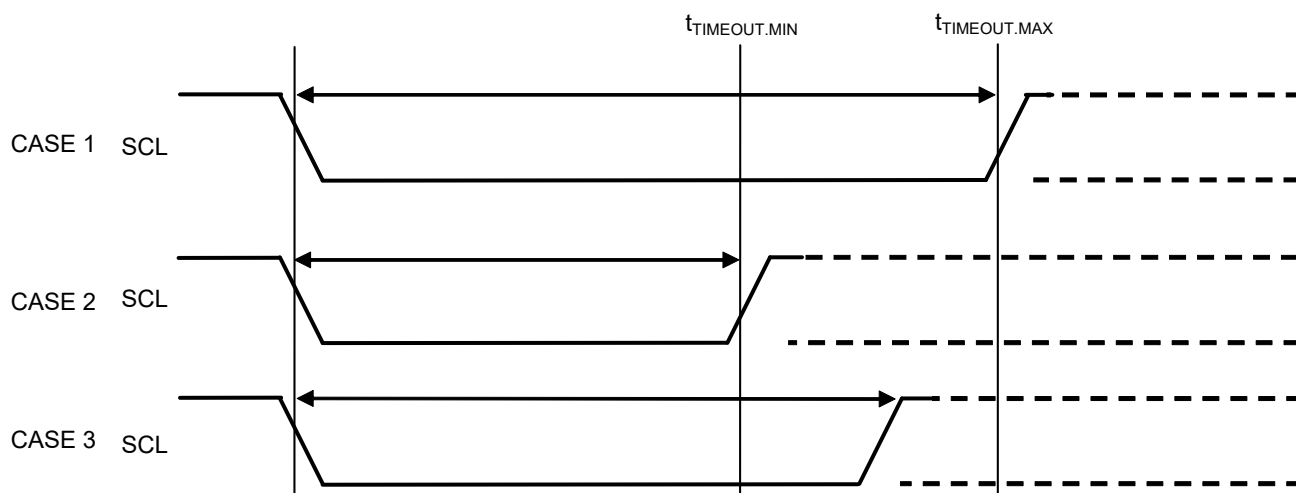
Table 12 Operating Modes

| Mode                                     | R/W | Byte | I/O Sequence                                                                           |
|------------------------------------------|-----|------|----------------------------------------------------------------------------------------|
| E <sup>2</sup> PROM current address read | 1   | 1    | Start, device select, R/W = 1, data, stop                                              |
| E <sup>2</sup> PROM random read          | 0   | 1    | Start, device select, R/W = 0, address                                                 |
|                                          | 1   |      | Restart, device select, R/W = 1, data, stop                                            |
| E <sup>2</sup> PROM sequential read      | 1   | ≥ 1  | Similar to E <sup>2</sup> PROM current address read or E <sup>2</sup> PROM random read |
| E <sup>2</sup> PROM byte write           | 0   | 1    | Start, device select, R/W = 0, address, data, stop                                     |
| E <sup>2</sup> PROM page write           | 0   | ≤ 16 | Start, device select, R/W = 0, address, data, stop                                     |
| SWPn and CWP                             | 0   | 2    | Start, device select, R/W = 0, X*1, X*1, stop                                          |
| Set E <sup>2</sup> PROM page address     | 0   | 2    | Start, device select, R/W = 0, X*1, X*1, stop                                          |
| Read E <sup>2</sup> PROM page address    | 1   | 2    | Start, device select, R/W = 1, X*1, X*1, stop                                          |
| Temperature sensor read                  | 1   | 2    | Start, device select, R/W = 1, data, stop                                              |
| Temperature sensor write                 | 0   | 2    | Start, device select, R/W = 0, pointer, data, stop                                     |

\*1. X: Don't care

## 10. Timeout

This IC has the timeout function. If the SCL stays "L" for the SCL clock "L" timeout ( $t_{\text{TIMEOUT}}$ ) or more, this IC resets the serial interface and returns to standby mode. If the SCL stays "L" for less than the  $t_{\text{TIMEOUT}}$ , this IC does not reset the serial interface. The  $t_{\text{TIMEOUT}}$  is 30 ms typ.



CASE 1: SCL clock time "L" ( $t_{\text{LOW}}$ )  $\geq t_{\text{TIMEOUT.MAX}}$ , this IC will reset the bus communication and return to standby mode.

CASE 2:  $t_{\text{LOW}} < t_{\text{TIMEOUT.MIN}}$ , this IC will not reset the bus communication.

CASE 3:  $t_{\text{TIMEOUT.MIN}} \leq t_{\text{LOW}} < t_{\text{TIMEOUT.MAX}}$ , this IC may or may not reset the bus communication.

**Figure 8 The Examples of the Timeout Timing**

When this IC is not in the range of the clock frequency specified by AC characteristics, it may not perform communication normally.

## ■ E<sup>2</sup>PROM Operation

### 1. Write

#### 1.1 Byte write

When this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "0", following a start condition, this IC generates an acknowledge.

This IC then receives an 8-bit word address and responds with an acknowledge. After this IC receives 8-bit write data and responds with an acknowledge, it receives a stop condition and that initiates the write cycle at the designated memory address.

When the certain word address is protected by the write instruction, this IC does not generate an acknowledge after data byte coding, and write operation is not performed.

During the write operation to this IC, access to the temperature sensor section of this IC are permitted, but all operations of the E<sup>2</sup>PROM section are inhibited to be performed and the E<sup>2</sup>PROM section does not send back an acknowledge.

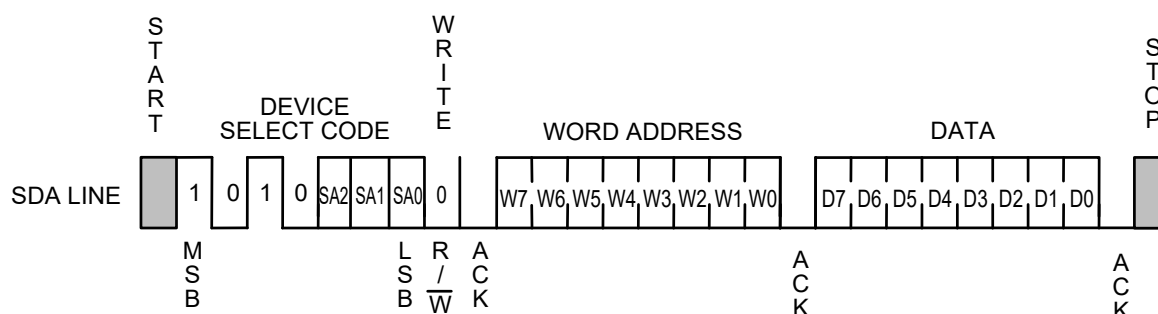


Figure 9 Byte Write

#### 1.2 Page write

The page write mode allows up to 16 bytes to be written in a single write operation in this IC.

Its basic process to transmit data is as same as byte write, but it operates page write by sequentially receiving 8-bit write data as much data as the page size has.

When this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "0", following a start condition, it generates an acknowledge. Then this IC receives an 8-bit word address, and responds with an acknowledge. After this IC receives 8-bit write data and responds with an acknowledge, it receives 8-bit write data corresponding to the next word address, and generates an acknowledge. This IC repeats reception of 8-bit write data and generation of acknowledge in succession. This IC can receive as many write data as the maximum page size.

Receiving a stop condition initiates a write cycle of the area starting from the designated memory address and having the page size equal to the received write data.

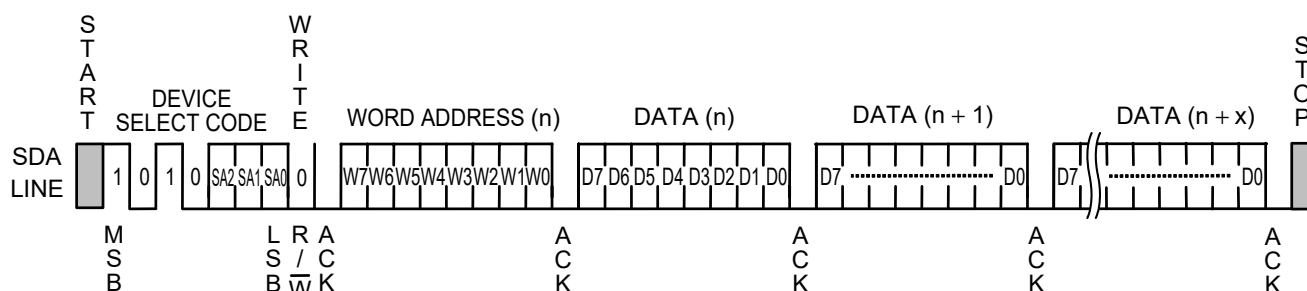


Figure 10 Page Write

The lower 4 bits of the word address are automatically incremented every time when it receives 8-bit write data. If the size of the write data exceeds 16 bytes, the higher 4 bits (W7 to W4) of the word address remain unchanged, and the lower 4 bits are rolled over and the last 16-byte data that this IC received will be overwritten.

### 1.3 Software write protect

This IC has set write protection for block n (SWPn), clear write protection for all blocks (CWP) and read protection status for block n (RPSn).

There are four independent memory blocks, and each block may be independently protected. The memory blocks are:

- Block 0 = word addresses 00h to 7Fh, page address = 0
- Block 1 = word addresses 80h to FFh, page address = 0
- Block 2 = word addresses 00h to 7Fh, page address = 1
- Block 3 = word addresses 80h to FFh, page address = 1

#### 1.3.1 Set write protect (SWPn) and clear write protect (CWP)

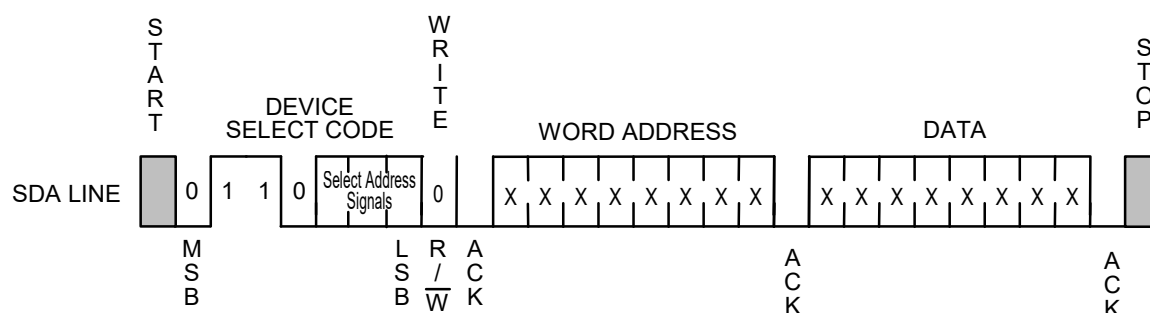
If the software write protect has been set with the SWPn instruction, the block n in memory is write-protected.

The four independent blocks are protected by SWPn instructions. The write-protected block can be cleared with the CWP instruction.

The CWP instruction clears write protection for all blocks, therefore the CWP instruction can not clear write protection for each block.

The SWPn and CWP instructions have the same format as a byte write instruction, but have a different device select code. Like the byte write instruction, it is followed by an address byte and a data byte, but in this case the contents can be set in all "Don't care". In the instructions of SWPn and CWP, be sure to apply the high voltage of V<sub>HV</sub> to the SA0 pin, and input "H" or "L" to the SA1 pin and SA2 pin.

The device select code for each block is shown in **Table 11**.



**Remark** X: Don't care

**Figure 11 Software Write Protect**

#### 1.3.2 Read protection status (RPSn)

The RPSn are the instructions to find the write protection status in block n. If a certain block is not protected by SWPn instruction, this IC generates an acknowledge after the device receives the device select code of the block. If a certain block is protected by SWPn instruction, this IC does not generate an acknowledge after the device receives the device select code of the block.

#### 1.3.3 Set E<sup>2</sup>PROM page address (SPAn)

The SPAn are the instructions to select the lower 256-byte page (SPA0) or the higher 256-byte page (SPA1). The E<sup>2</sup>PROM page address selects the lower 256 bytes (SPA0) after power-on reset.

#### 1.3.4 Read E<sup>2</sup>PROM page address (RPA)

The RPA are the instructions to find the current page address status. If the current page address is "0", this IC generates an acknowledge after the device receives the device select code. If the current page address is "1", this IC does not generate an acknowledge.

**Table 13 Acknowledge for Write Instruction (R/W bit = 0)**

| Status                        | Instruction                                    | ACK Output | Word Address | ACK Output | Data       | ACK Output | Write |
|-------------------------------|------------------------------------------------|------------|--------------|------------|------------|------------|-------|
| Software Write Protect (SWPn) | SWPn in protected block                        | No         | Don't care   | No         | Don't care | No         | No    |
|                               | SWPn in no protected block                     | Yes        | Don't care   | Yes        | Don't care | Yes        | Yes   |
|                               | CWP                                            | Yes        | Don't care   | Yes        | Don't care | Yes        | Yes   |
|                               | Page write or byte write in protected block    | Yes        | Word address | Yes        | Don't care | No         | No    |
|                               | Page write or byte write in no protected block | Yes        | Word address | Yes        | Data       | Yes        | Yes   |
| No Software Write Protect     | SWPn or CWP                                    | Yes        | Don't care   | Yes        | Don't care | Yes        | Yes   |
|                               | Page write or byte write                       | Yes        | Word address | Yes        | Data       | Yes        | Yes   |

**Table 14 Acknowledge for Read Instruction (R/W bit = 1)**

| Status                        | Instruction | ACK Output | Word Address | ACK Output | Data       | ACK Output |
|-------------------------------|-------------|------------|--------------|------------|------------|------------|
| Software Write Protect (SWPn) | RPSn        | No         | Don't care   | No         | Don't care | No         |
| No Software Write Protect     | RPSn        | Yes        | Don't care   | No         | Don't care | No         |

#### 1.4 Acknowledge polling

Acknowledge polling is used to know the completion of the write cycle in this IC.

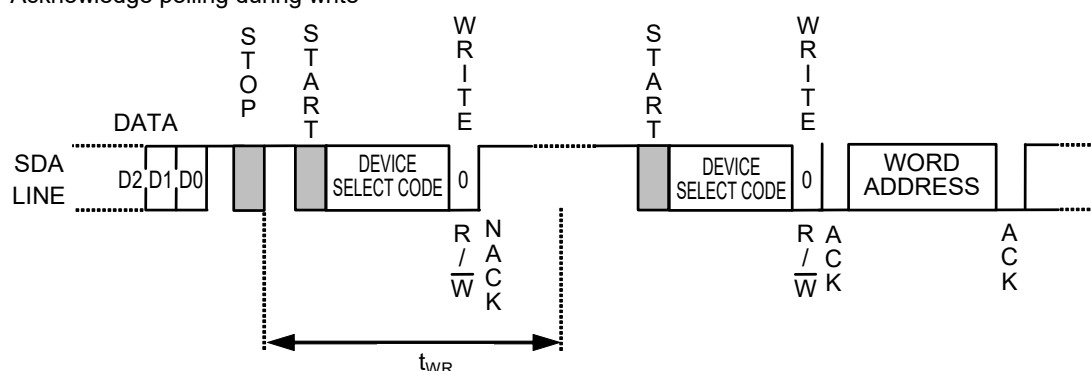
After this IC receives a stop condition and once starts the write cycle, E<sup>2</sup>PROM operations are forbidden and no response is made to the signal transmitted by the master device.

Accordingly the master device can recognize the completion of the write cycle in this IC (slave device) by detecting a response from the slave device after transmitting the start condition, the device address and the read / write instruction code to this IC, namely to the slave devices.

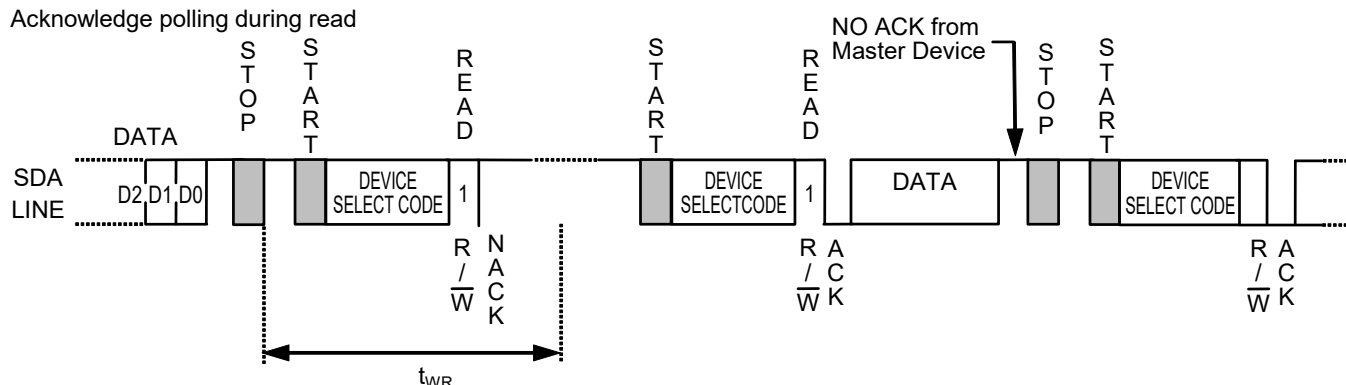
That is, if this IC does not generate an acknowledge, the write cycle is in progress and if this IC generates an acknowledge, the write cycle has been completed.

It is recommended to use the read instruction "1" as the read / write instruction code transmitted by the master device.

Acknowledge polling during write



Acknowledge polling during read



**Remark** Users are able to input data after acknowledge output in acknowledge polling during write.  
Users are able to read data after acknowledge output in acknowledge polling during read.  
However, after that users input the write instruction, a start condition may not be input during data output. Input a stop condition and the next instruction after acknowledge output and data output.

Figure 12 Usage Example of Acknowledge Polling

## 2. Read

### 2.1 Current address read

Either in writing or in reading this IC holds the last accessed memory address. The memory address is maintained when the instruction transmission is not interrupted, and the memory address is maintained as long as the power voltage does not decrease less than the power-on reset threshold voltage ( $V_{PON}$ ).

The master device can read the data at the memory address of the current address pointer without assigning the word address as a result, when it recognizes the position of the address pointer in this IC. This is called "current address read".

In the following the address counter in this IC is assumed to be "n".

When this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "1" following a start condition, it responds with an acknowledge.

Next an 8-bit data at the address "n" is sent from this IC synchronous to the SCL clock. The address counter is incremented and the content of the address counter becomes  $n + 1$ . The master device outputs stop condition not an acknowledge, the reading of this IC is ended.

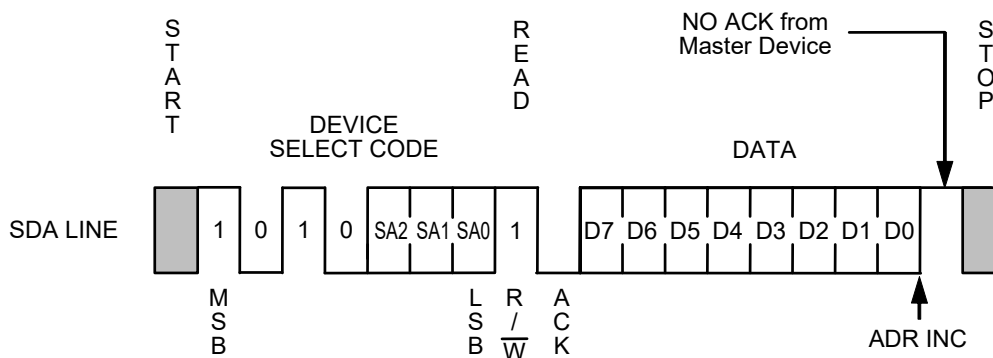


Figure 13 Current Address Read

Attention should be paid to the following point on the recognition of the address pointer in this IC.

In read, the memory address counter in this IC is automatically incremented after output of the 8th bit of the data. In write, on the other hand, the higher bits of the memory address (the higher 4 bits of the word address) are left unchanged and are not incremented.

## 2.2 Random read

Random read is used to read the data at an arbitrary memory address.

A dummy write is performed to load the memory address into the address counter.

When this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "0" following a start condition, it responds with an acknowledge.

This IC then receives an 8-bit word address and responds with an acknowledge. The memory address is loaded to the address counter in this IC by these operations. Reception of write data does not follow in a dummy write whereas reception of write data follows in byte write and in page write.

Since the memory address is loaded into the memory address counter by dummy write, the master device can read the data starting from the arbitrary memory address by transmitting a new start condition and performing the same operation in the current address read.

That is, when this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "1", following a start condition signal, it responds with an acknowledge. Next, 8-bit data is transmitted from this IC synchronously with the SCL clock. The master device outputs stop condition not an acknowledge, the reading of this IC is ended.

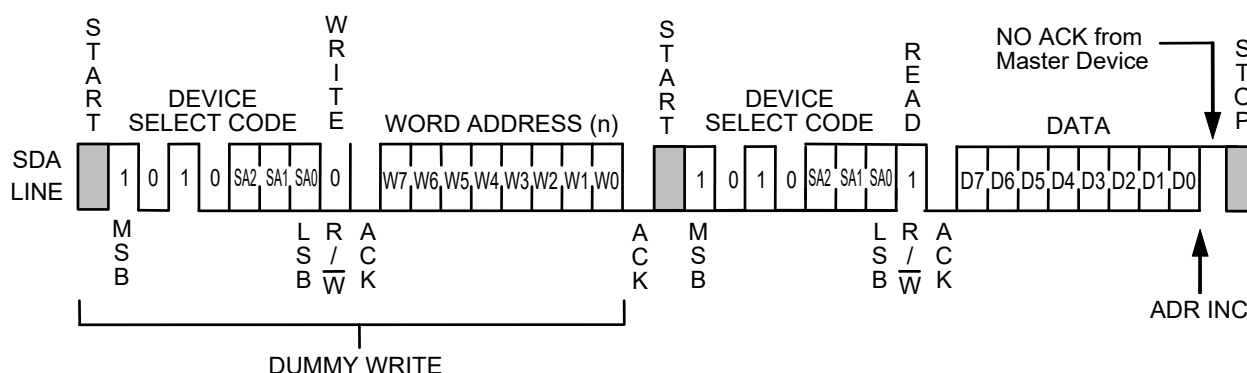


Figure 14 Random Read

## 2.3 Sequential read

When this IC receives a 7-bit device address and a 1-bit read / write instruction code set to "1" following a start condition both in current address read and random read, it responds with an acknowledge.

When an 8-bit data is output from this IC synchronously with the SCL clock, the address counter is automatically incremented.

When the master device responds with an acknowledge, the data at the next memory address is transmitted. Response with an acknowledge by the master device has the memory address counter in this IC incremented and makes it possible to read data in succession. This is called sequential read.

The master device outputs stop condition not an acknowledge, the reading of this IC is ended.

Data can be read in succession in the sequential read mode. When the memory address counter reaches the last word address, it rolls over to the first word address of same page address.

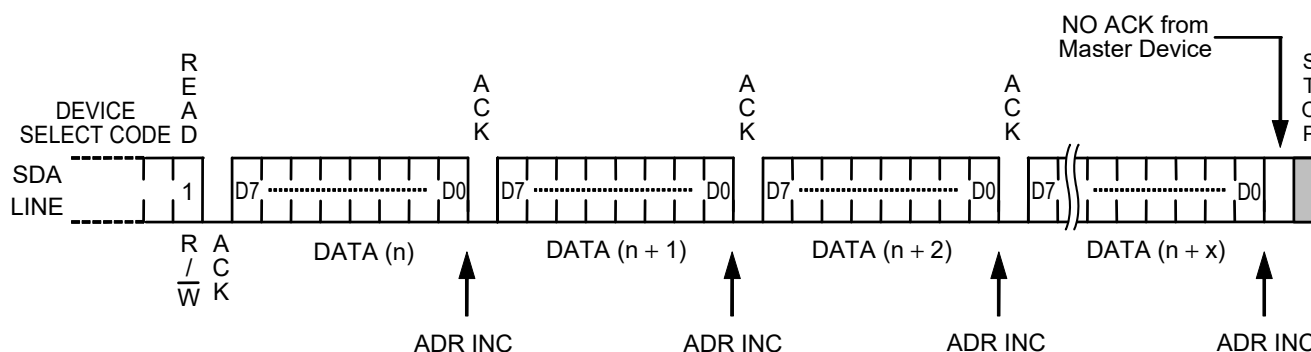


Figure 15 Sequential Read

## ■ Temperature Sensor Operation

Temperature sensor goes into active condition and regularly refresh ambient temperature register value by ambient temperature and the comparison value between ambient temperature and temperature limit after power-on.

The Temperature registers store the ambient temperature data, limits, and configuration values. All registers in the address space from "00h" through "08h" are 16-bit registers, accessed through block read and write commands.

Temperature sensor goes into shutdown condition by setting SHDN bit of configuration register.

Temperature sensor circuit including temperature sensor device and A/D converter circuit are stopped to control the current consumption during shutdown mode. Ambient temperature register value is held during shutdown condition.

Temperature sensor goes into active condition again by clearing SHDN bit of configuration register.

### 1. Temperature sensor register write operation

Writing to the temperature registers are accomplished through continuous write operations for 2-byte data. After the write operation of the device select code, the master device writes data to the pointer register, then it writes the 16-bit data every 8 bits. The 16-bit data is stored internally at the time when this IC generates an acknowledge after receiving the 16-bit data.

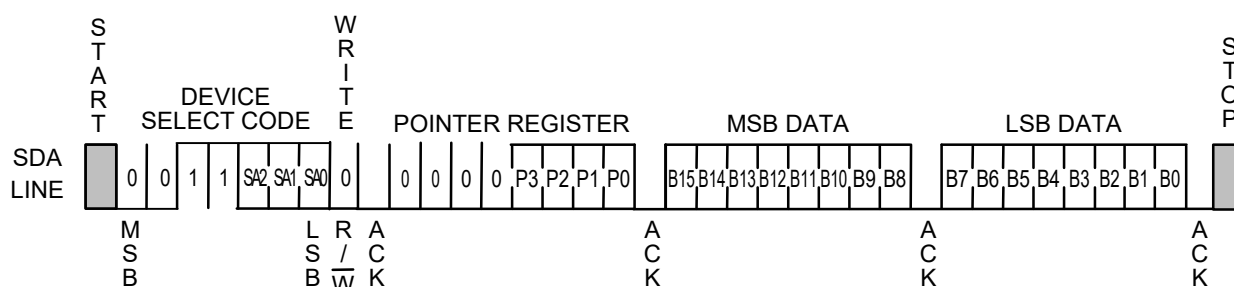


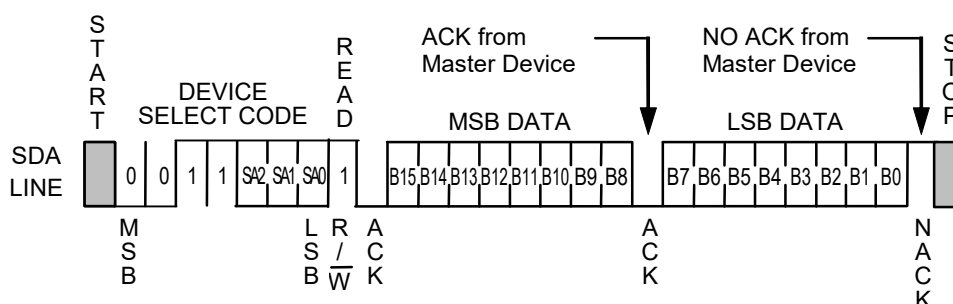
Figure 16 Temperature Sensor Register Write

## 2. Temperature sensor register read operation

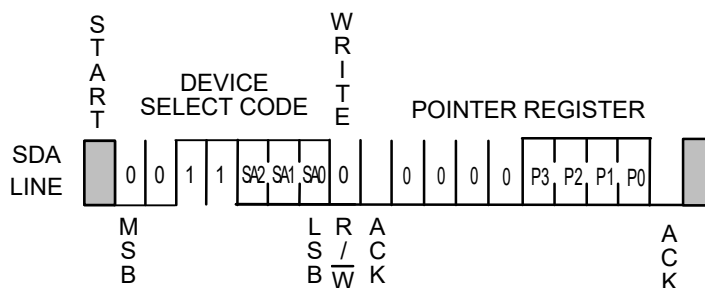
Reading data from temperature sensor register may be accomplished in the following ways:

- (1) If the location latched in the pointer register is correct, the read sequence may consist of an operation in which R/W bit of device select code is set to "1", and a following operation in which the 2-byte data is output every 8 bits from this IC.  
 This way is used to read data from the ambient temperature register repeatedly.
- (2) The master device transfers the device select code and the R/W bit "0" and it is followed by the pointer register. The current location latched in the pointer register becomes clear by sending stop condition after this IC generates an acknowledge.  
 After that, reading data from the temperature sensor will be accomplished in the way of (1).

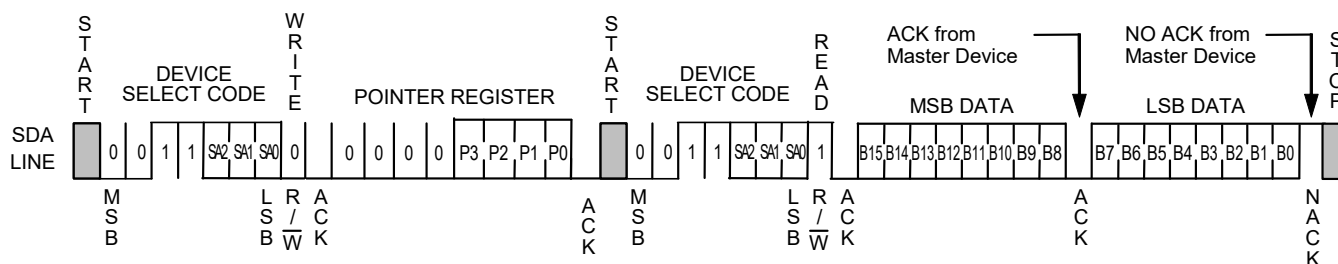
The master device is able to transfer the reading instruction by sending start condition instead of stop condition after the pointer register is set in the way of (2).



**Figure 17 Temperature Sensor Register Read**



**Figure 18 Pointer Register Write**



**Figure 19 Pointer Register Write and Temperature Sensor Register Read**

### 3. Temperature sensor register

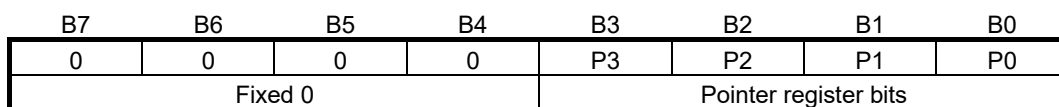
The register addresses are shown in **Table 15**.

**Table 15 Pointer Register Mapping**

| Address        | R/W               | Name                         | Function                                                           | Default        |
|----------------|-------------------|------------------------------|--------------------------------------------------------------------|----------------|
| Not applicable | W                 | Address pointer              | Address storage for subsequent operations                          | Undefined      |
| 00h            | R                 | Capabilities register        | Indicates the functions and capabilities of the temperature sensor | 00EFh          |
| 01h            | R/W <sup>*1</sup> | Configuration register       | Controls the operation of the temperature sensor                   | 0000h          |
| 02h            | R/W               | HIGH limit register          | Temperature high limit                                             | 0000h          |
| 03h            | R/W               | LOW limit register           | Temperature low limit                                              | 0000h          |
| 04h            | R/W               | TCRIT limit register         | Critical temperature                                               | 0000h          |
| 05h            | R                 | Ambient temperature register | Current ambient temperature                                        | Not applicable |
| 06h            | R                 | Manufacturer ID register     | PCI-SIG manufacturer ID                                            | 1C85h          |
| 07h            | R                 | Device / revision register   | Device ID and revision number                                      | 2221h          |
| 08h            | R/W               | Resolution register          | Sets temperature resolution                                        | 0001h          |
| 09h to FFh     | – <sup>*2</sup>   | Undefined register           | Undefined                                                          | Undefined      |

<sup>\*1</sup>. There are read-only bits, write-only bits, and readable and writable bits in configuration register.

<sup>\*2</sup>. Behavior on accesses to invalid pointer register locations is not assured.



**Figure 20 Pointer Register Structure**

### 3.1 Capabilities register

The capabilities register indicates the supported features of the temperature sensor.

|      |       |     |           |     |       |     |       |
|------|-------|-----|-----------|-----|-------|-----|-------|
| B15  | B14   | B13 | B12       | B11 | B10   | B9  | B8    |
| RFU  | RFU   | RFU | RFU       | RFU | RFU   | RFU | RFU   |
| B7   | B6    | B5  | B4        | B3  | B2    | B1  | B0    |
| EVSD | TMOUT | VHV | TRES[1:0] |     | RANGE | ACC | EVENT |

Address : 00h

R/W : Read only

Default : 00EFh

**Figure 21 Capabilities Register Structure**

#### Bit 15 to 8 : RFU

Reserved for future use. These bits are always "0".

#### Bit 7 : EVSD

$\overline{\text{EVENT}}$  pin operation during shutdown mode.

"1":  $\overline{\text{EVENT}}$  pin output de-asserts during shutdown.

#### Bit 6 : TMOUT

Bus timeout period access during normal operation.

"1":  $t_{\text{TIMEOUT}}$  is supported within the range of 25 ms to 35 ms.

#### Bit 5 : VHV

Support SA0 high level input voltage ( $V_{\text{HV}}$ ) for SA0 pin.

"1": A voltage up to 10 V is supported on the SA0 pin.

#### Bit 4 to 3 : TRES[1:0]

Indicates the temperature resolution. Temperature resolution is set on the resolution register. Refer to "3.7 Resolution Register" for resolution register.

"00": 9-bit temperature resolution (0.5°C resolution)

"01" (default): 10-bit temperature resolution (0.25°C resolution)

"10": 11-bit temperature resolution (0.125°C resolution)

"11": 12-bit temperature resolution (0.0625°C resolution)

#### Bit 2 : RANGE

Indicates the supported temperature range.

"1": Lower than 0°C can be read and the sign bit will be set appropriately.

#### Bit 1 : ACC

Indicates the supported temperature accuracy.

"1":  $\pm 1.0^\circ\text{C}$  max. (+75°C to +95°C)

#### Bit 0 : EVENT

Indicates whether interrupt mode is supported.

"1": Interrupt mode is supported.

### 3.2 Configuration register

The configuration register holds the status bits and setting data of the  $\overline{\text{EVENT}}$  pin, as well as temperature hysteresis width setting data.

|               |               |       |              |               |               |              |               |
|---------------|---------------|-------|--------------|---------------|---------------|--------------|---------------|
| B15           | B14           | B13   | B12          | B11           | B10           | B9           | B8            |
| RFU           | RFU           | RFU   | RFU          | RFU           | HYST[1:0]     |              | SHDN          |
| B7            | B6            | B5    | B4           | B3            | B2            | B1           | B0            |
| TCRIT<br>LOCK | EVENT<br>LOCK | CLEAR | EVENT<br>STS | EVENT<br>CTRL | TCRIT<br>ONLY | EVENT<br>POL | EVENT<br>MODE |

Address : 01h

R/W : Readable and Writable

Default : 0000h

**Figure 22 Configuration Register Structure**

#### Bit 15 to 11 : RFU

Reserved for future use. These bits are always "0".

#### Bit 10 to 9: HYST[1:0]

Control the hysteresis width shown in **Table 16**. This hysteresis width applies to temperature high limit, temperature low limit and critical temperature. Once the ambient temperature is above a given threshold, it must drop below the threshold minus the hysteresis width in order to be flagged as an interrupt event.

Note that hysteresis width is also applied to the  $\overline{\text{EVENT}}$  pin functionality. When either of the lock bits is set, these bits HYST[1:0] cannot be altered.

**Table 16 Hysteresis Width Setting**

| HYST1 | HYST0 | Hysteresis Width |
|-------|-------|------------------|
| 0     | 0     | No hysteresis    |
| 0     | 1     | 1.5°C            |
| 1     | 0     | 3.0°C            |
| 1     | 1     | 6.0°C            |

#### Bit 8 : SHDN

Shutdown setting. The temperature sensing device and A/D converter are disabled to save power, no events will be generated. When either of the lock bits is set, this bit cannot be set until unlocked. However it can be cleared at any time.

"0" (default): The temperature sensor is active and converting.

"1": The temperature sensor is disabled and will not generate interrupts or update the temperature data.

#### Bit 7 : TCRIT\_LOCK

Locks the TCRIT limit register from being updated.

"0" (default): The TCRIT limit register can be updated normally.

"1": The TCRIT limit register is locked and cannot be updated. Once this bit has been set, it cannot be cleared until an internal power-on reset.

#### Bit 6 : EVENT\_LOCK

Locks the HIGH and LOW limit registers from being updated.

"0" (default): The HIGH and LOW limit registers can be updated normally.

"1": The HIGH and LOW limit registers are locked and cannot be updated. Once this bit has been set, it cannot be cleared until an internal power-on reset.

**Bit 5 : CLEAR**

Clears the  $\overline{\text{EVENT}}$  pin when it has been asserted. This bit is write-only and will always be read "0".

"0": Nothing happens.

"1": In interrupt mode, the  $\overline{\text{EVENT}}$  pin is reset and will not be asserted until a new interrupt condition occurs. This bit is ignored if the device is operating in comparator mode and when detecting the critical temperature. This bit is self clearing. The read value is fixed to "0".

**Bit 4 : EVENT\_STS**

Indicates if the  $\overline{\text{EVENT}}$  pin is asserted. This bit is read only.

"0" (default): The  $\overline{\text{EVENT}}$  pin is not being asserted.

"1": The  $\overline{\text{EVENT}}$  pin is being asserted.

**Bit 3 : EVENT\_CTRL**

Controls the asserted state of the  $\overline{\text{EVENT}}$  pin. If either of the lock bits are set (bit 7 and bit 6), then this bit cannot be altered.

"0" (default): The  $\overline{\text{EVENT}}$  pin can not be asserted.

"1": The  $\overline{\text{EVENT}}$  pin can be asserted.

**Bit 2 : TCRIT\_ONLY**

Controls whether the  $\overline{\text{EVENT}}$  pin will be asserted from a high / low out-of-limit condition. When the EVENT\_LOCK bit is set, this bit cannot be altered.

"0" (default): The  $\overline{\text{EVENT}}$  pin will be asserted if the measured temperature is above the temperature high limit or below the temperature low limit in addition to if the temperature is above the critical temperature.

"1": The  $\overline{\text{EVENT}}$  pin will only be asserted if the measured temperature is above the critical temperature.

**Bit 1 : EVENT\_POL**

Controls the active "H" state of the  $\overline{\text{EVENT}}$  pin. The  $\overline{\text{EVENT}}$  pin is driven to this state when it is asserted. If either of the lock bits are set (bit 7 and bit 6), then this bit cannot be altered.

"0" (default): The  $\overline{\text{EVENT}}$  pin is active "L". The active "L" state of the pin will be logical "0".

"1": The  $\overline{\text{EVENT}}$  pin is active "H". The active "H" state of the pin will be logical "1".

**Bit 0 : EVENT\_MODE**

Controls the behavior of the  $\overline{\text{EVENT}}$  pin. The  $\overline{\text{EVENT}}$  pin may function in either comparator or interrupt mode. If either of the lock bits are set (bit 7 and bit 6), then this bit cannot be altered.

"0" (default): The  $\overline{\text{EVENT}}$  pin will function in comparator mode.

"1": The  $\overline{\text{EVENT}}$  pin will function in interrupt mode.

### 3.3 Temperature limit register (HIGH limit / LOW limit / TCRIT limit)

Temperature in the HIGH limit register, LOW limit register and TCRIT limit register are set and described in 2's complement format of Sign bit + 10-bit.

Examples of temperature limit register are shown in **Table 17**.

| B15               | B14               | B13               | B12               | B11                | B10                | B9                | B8                |
|-------------------|-------------------|-------------------|-------------------|--------------------|--------------------|-------------------|-------------------|
| –                 | –                 | –                 | Sign              | 2 <sup>7</sup> °C  | 2 <sup>6</sup> °C  | 2 <sup>5</sup> °C | 2 <sup>4</sup> °C |
| B7                | B6                | B5                | B4                | B3                 | B2                 | B1                | B0                |
| 2 <sup>3</sup> °C | 2 <sup>2</sup> °C | 2 <sup>1</sup> °C | 2 <sup>0</sup> °C | 2 <sup>-1</sup> °C | 2 <sup>-2</sup> °C | –                 | –                 |

HIGH limit register

Address : 02h

R/W : Readable and writable

Default : 0000h

LOW limit register

Address : 03h

R/W : Readable and writable

Default : 0000h

TCRIT limit register

Address : 04h

R/W : Readable and writable

Default : 0000h

**Figure 23 Temperature Limit Register Structure**

### 3.4 Ambient temperature register

3 status bits of TCRIT, HIGH and LOW and ambient temperature data described in 2's complement format of Sign bit + 12-bit can be read.

Examples of ambient temperature register are shown in **Table 17**.

|                   |                   |                   |                   |                    |                      |                      |                      |
|-------------------|-------------------|-------------------|-------------------|--------------------|----------------------|----------------------|----------------------|
| B15               | B14               | B13               | B12               | B11                | B10                  | B9                   | B8                   |
| TCRIT             | HIGH              | LOW               | Sign              | 2 <sup>7</sup> °C  | 2 <sup>6</sup> °C    | 2 <sup>5</sup> °C    | 2 <sup>4</sup> °C    |
| B7                | B6                | B5                | B4                | B3                 | B2                   | B1                   | B0                   |
| 2 <sup>3</sup> °C | 2 <sup>2</sup> °C | 2 <sup>1</sup> °C | 2 <sup>0</sup> °C | 2 <sup>-1</sup> °C | 2 <sup>-2</sup> °C*1 | 2 <sup>-3</sup> °C*1 | 2 <sup>-4</sup> °C*1 |

Address : 05h

R/W : Read-only

Default : Not applicable (0000h)

- \*1. The bits may not be used according to the resolution defined based on value of TRES[1:0] bit of the capabilities register. Unused/unsupported bits will read as 0.

**Figure 24 Ambient Temperature Register Structure**

#### Bit 15 : TCRIT

When set, the ambient temperature is above the critical temperature. This bit will remain set so long as the ambient temperature is above the critical temperature and will automatically clear once the ambient temperature has dropped below the limit minus the hysteresis width.

#### Bit 14 : HIGH

When set, the ambient temperature is above the high limit. This bit will remain set so long as the ambient temperature is above the high limit. Once set, it will only be cleared when the ambient temperature drops below or equal to the high limit minus the hysteresis width.

#### Bit 13 : LOW

When set, the ambient temperature is below the low limit. This bit will remain set so long as the ambient temperature is below the low limit minus the hysteresis. Once set, it will only be cleared when the ambient temperature meets or exceeds the low limit.

#### Bit 12 to 0 : TAMB

Ambient temperature data in 2's complement format of Sign bit + 12-bit.

**Table 17 Examples of Temperature**

| B15 to B0 (binary)  | Value   | Unit |
|---------------------|---------|------|
| xxx0 0111 1101 00xx | +125.00 | °C   |
| xxx0 0101 0101 00xx | +85.00  | °C   |
| xxx0 0001 1001 00xx | +25.00  | °C   |
| xxx0 0000 0010 11xx | +2.75   | °C   |
| xxx0 0000 0001 00xx | +1.00   | °C   |
| xxx0 0000 0000 01xx | +0.25   | °C   |
| xxx0 0000 0000 00xx | +0.00   | °C   |
| xxx1 1111 1111 11xx | -0.25   | °C   |
| xxx1 1111 1111 00xx | -1.00   | °C   |
| xxx1 1111 1101 01xx | -2.75   | °C   |
| xxx1 1110 1100 00xx | -20.00  | °C   |

### 3.5 Manufacturer ID register

PCI-SIG ID number of ABLIC Inc. can be read.

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 |
| 0   | 0   | 0   | 1   | 1   | 1   | 0  | 0  |

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
| 1  | 0  | 0  | 0  | 0  | 1  | 0  | 1  |

Address : 06h

R/W : Read-only

Default : 1C85h

Figure 25 Manufacturer ID Register Structure

### 3.6 Device ID / revision register

Device ID and revision number of product can be read.

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 |
| 0   | 0   | 1   | 0   | 0   | 0   | 1  | 0  |

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 |
| 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  |

Address : 07h

R/W : Read-only

Default : 2221h

Figure 26 Device ID / Revision Register Structure

### 3.7 Resolution register

Resolution register defines temperature sensor resolution.

|     |     |     |     |     |     |    |    |
|-----|-----|-----|-----|-----|-----|----|----|
| B15 | B14 | B13 | B12 | B11 | B10 | B9 | B8 |
| 0   | 0   | 0   | 0   | 0   | 0   | 0  | 0  |

|    |    |    |    |    |    |          |    |
|----|----|----|----|----|----|----------|----|
| B7 | B6 | B5 | B4 | B3 | B2 | B1       | B0 |
| 0  | 0  | 0  | 0  | 0  | 0  | RES[1:0] |    |

Address : 08h

R/W : Readable and writable

Default : 0001h

Figure 27 Resolution Register Structure

#### Bit 15 to 2 : –

Unimplemented bit. These bits are always "0".

#### Bit 1 to 0 : RES[1:0]

These bits define resolution.

The resolution here is reflected by TRES[1:0] value of capacities register.

"00": LSB = 0.5°C ( $t_{CONV} \leq 30$  ms max.)

"01" (default): LSB = 0.25°C ( $t_{CONV} \leq 60$  ms max.)

"10": LSB = 0.125°C ( $t_{CONV} \leq 125$  ms max.)

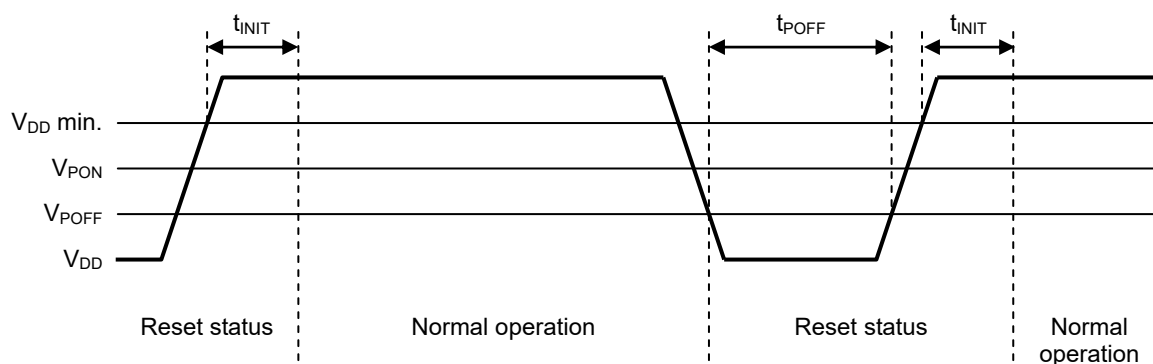
"11": LSB = 0.0625°C ( $t_{CONV} \leq 125$  ms max.)

## ■ Reset and Initialization

This IC has a power-on reset circuit which prevents malfunction, cancels write at power-on.

At power-on status, the master device should not transfer the data from the time that the power supply voltage reaches  $V_{DD \text{ min.}}$  to  $t_{INIT}$ . And at power-off, all registers are reset when the power supply voltage drops below  $V_{POFF}$ . Therefore if the power supply voltage remains below  $V_{POFF}$  for  $t_{POFF}$ , all operations of this IC are reset.

Operating this IC will be required to apply the stable power supply voltage ( $V_{DD}$ ). The power supply voltage must remain stable until the end of the transmission of the data and, for a write instruction and temperature conversion time.



**Figure 28** Operation when Power Supply Voltage Drops and Power-on

## ■ Usage

### 1. A pull-up resistor to SDA I/O pin and SCL input pin

In consideration of I<sup>2</sup>C-bus protocol function, the SDA I/O pin and SCL input pin should be connected with a pull-up resistor. This IC cannot transmit normally without using a pull-up resistor.

In case that the SCL input pin of this IC is connected to the Nch open-drain output pin of the master device, connect the SCL pin with a pull-up resistor. As well, in case the SCL input pin of this IC is connected to the tri-state output pin of the master device, connect the SCL pin with a pull-up resistor in order not to set it in "High-Z". This prevents this IC from error caused by an uncertain output (High-Z) from the tri-state pin when resetting the master device during the voltage drop.

Pull-up resistor of the  $\overline{\text{EVENT}}$  pin is also required to transmit  $\overline{\text{EVENT}}$  behavior.

### 2. Equivalent circuits of input pin and I/O pin

The SCL pin and the SDA pin of this IC do not have a built-in pull-down or pull-up resistor. Each of the SA0 pin, SA1 pin and SA2 pin has a built-in pull-down resistor. The SDA pin is an open-drain output. The followings are equivalent circuits of the pins.

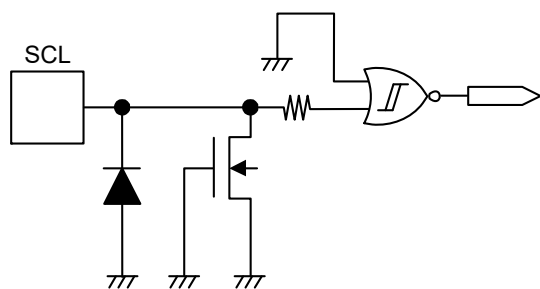


Figure 29 SCL Pin

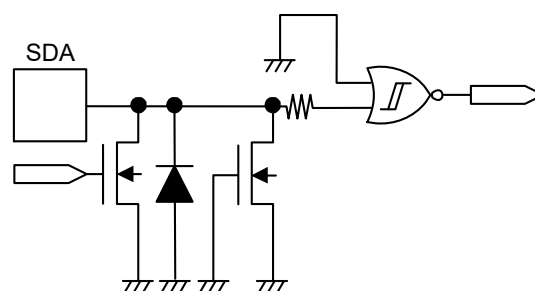


Figure 30 SDA Pin

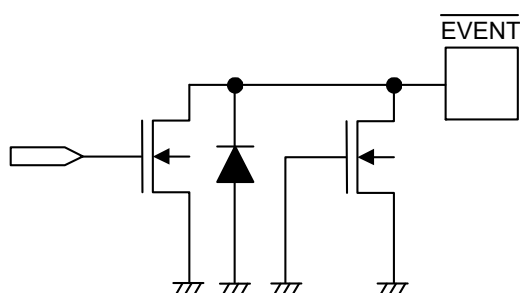


Figure 31  $\overline{\text{EVENT}}$  Pin

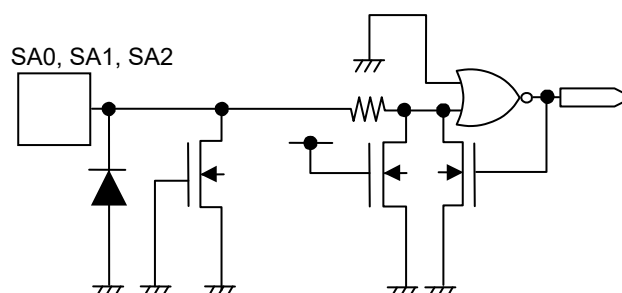


Figure 32 SA0, SA1, SA2 Pin

### 3. Acknowledge check

The I<sup>2</sup>C-bus protocol includes an acknowledge check function as a handshake function to prevent a communication error. This function allows detection of a communication failure during data communication between the master device and this IC. This function is effective to prevent malfunction, so it is recommended to perform an acknowledge check with the master device.

#### 4. SDA pin and SCL pin noise suppression time

This IC includes a built-in low-pass filter at the SDA pin and the SCL pin to suppress noise. If the power supply voltage is 2.2 V, noise with a pulse width of 75 ns or less can be suppressed.

For details of the assurable value, refer to noise suppression time ( $t_i$ ) in **Table 8** in "■ AC Electrical Characteristics".

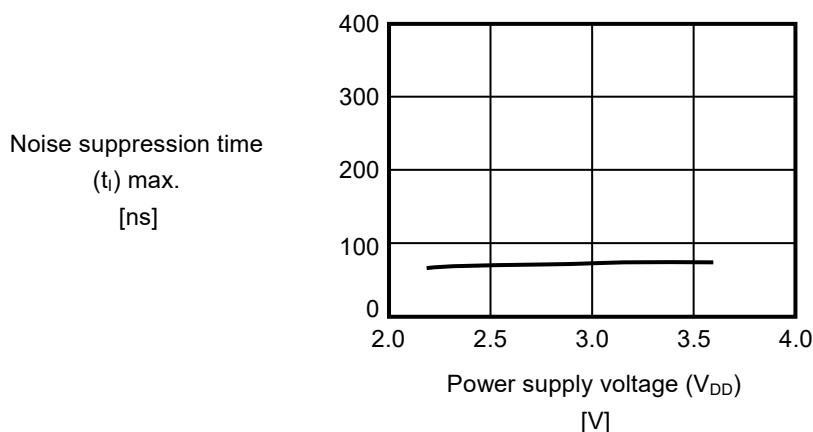


Figure 33 Noise Suppression Time for SDA Pin and SCL Pin

#### 5. Operation when inputting stop condition during write

This IC does the E<sup>2</sup>PROM write operation only when it receives data of 1 byte or more and receives a stop condition immediately after an acknowledge output.

Refer to **Figure 34** for details.

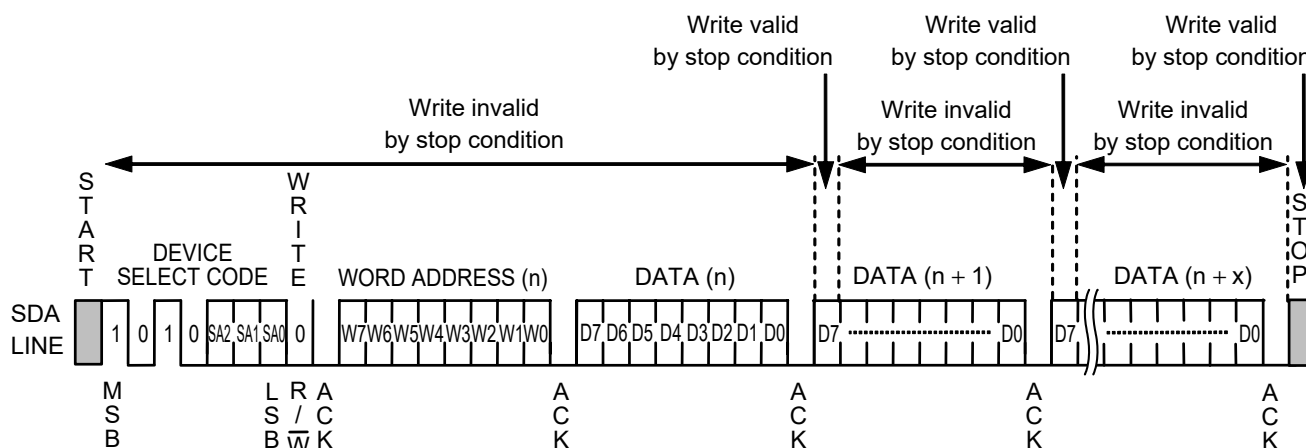


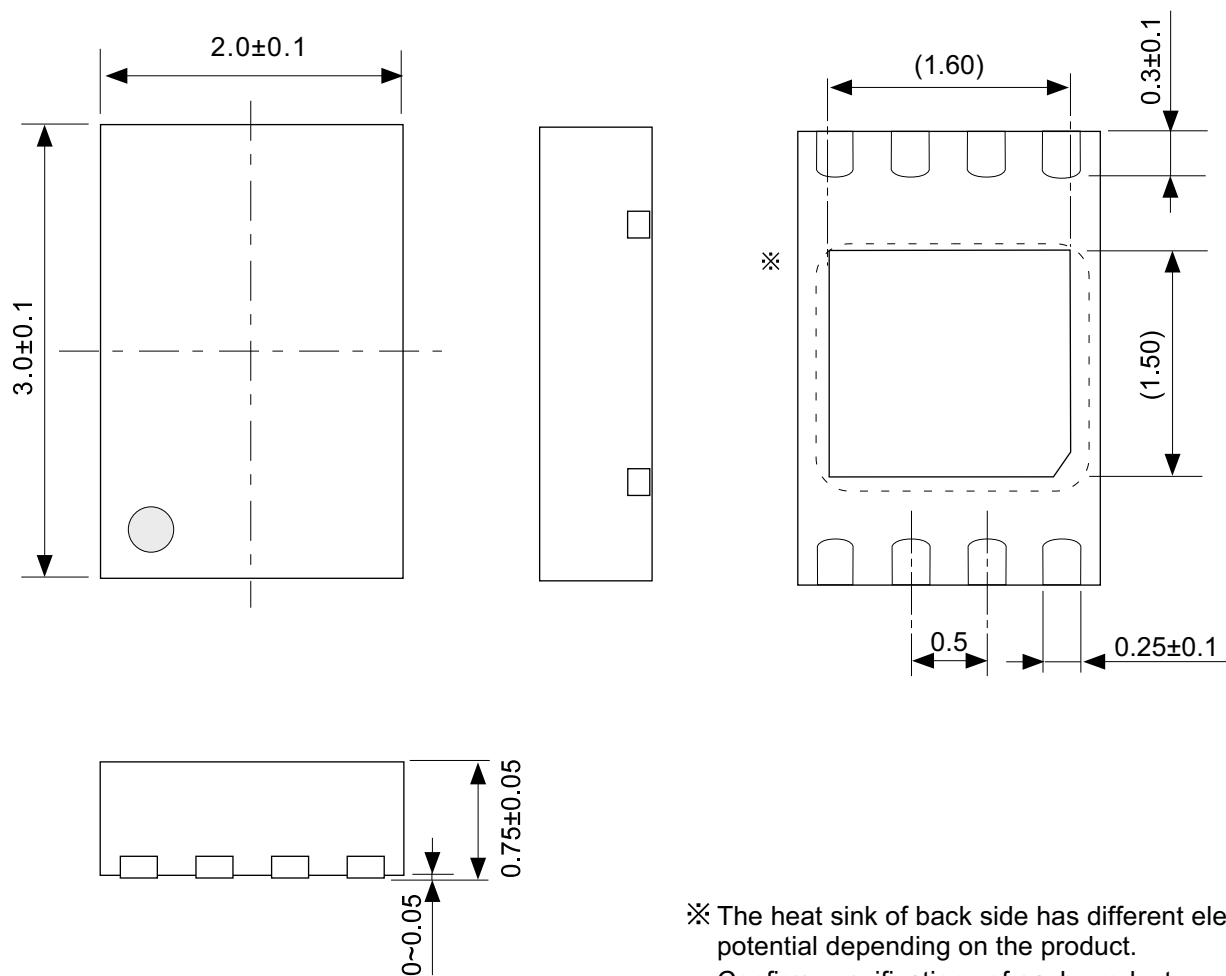
Figure 34 Write Operation by Inputting Stop Condition during Write

#### 6. Command cancel by start condition

By a start condition, users are able to cancel command which is being input. However, users are not able to input a start condition when this IC is outputting "L". When users cancel the command, there may be a case that the address will not be identified. Use random read for the read operation, not current address read.

## ■ Precautions

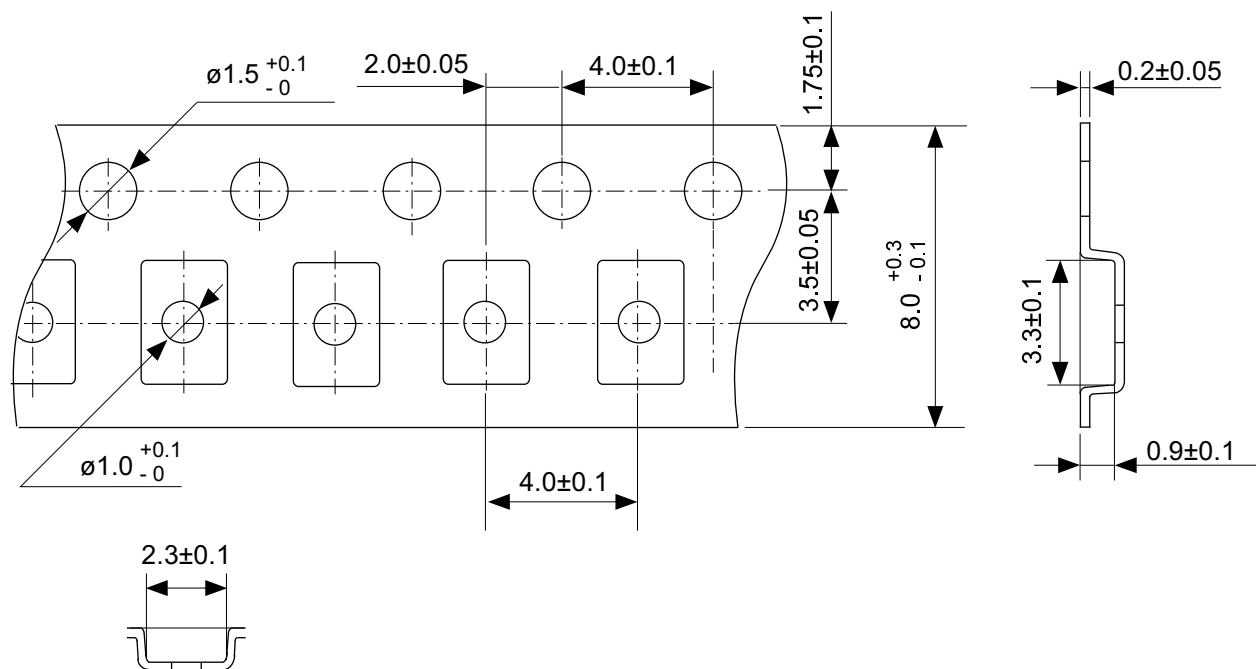
- Do not operate these ICs in excess of the absolute maximum ratings. Attention should be paid to the power supply voltage, especially. The surge voltage which exceeds the absolute maximum ratings can cause latch-up and malfunction. Perform operations after confirming the detailed operation condition in the data sheet.
- Operations with moisture on this IC's pins may occur malfunction by short-circuit between pins. Especially, in occasions like picking this IC up from low temperature tank during the evaluation. Be sure that not remain frost on this IC's pin to prevent malfunction by short-circuit.  
Also attention should be paid in using on environment, which is easy to dew for the same reason.
- Do not apply an electrostatic discharge to this IC that exceeds the performance ratings of the built-in electrostatic protection circuit.
- ABLIC Inc. claims no responsibility for any and all disputes arising out of or in connection with any infringement of the products including this IC upon patents owned by a third party.



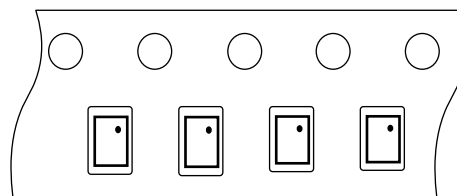
※ The heat sink of back side has different electric potential depending on the product.  
Confirm specifications of each product.  
Do not use it as the function of electrode.

No. PQ008-B-P-SD-1.0

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| TITLE      | DFN-8-B-PKG Dimensions                                                                |
| No.        | PQ008-B-P-SD-1.0                                                                      |
| ANGLE      |  |
| UNIT       | mm                                                                                    |
|            |                                                                                       |
| ABLIC Inc. |                                                                                       |



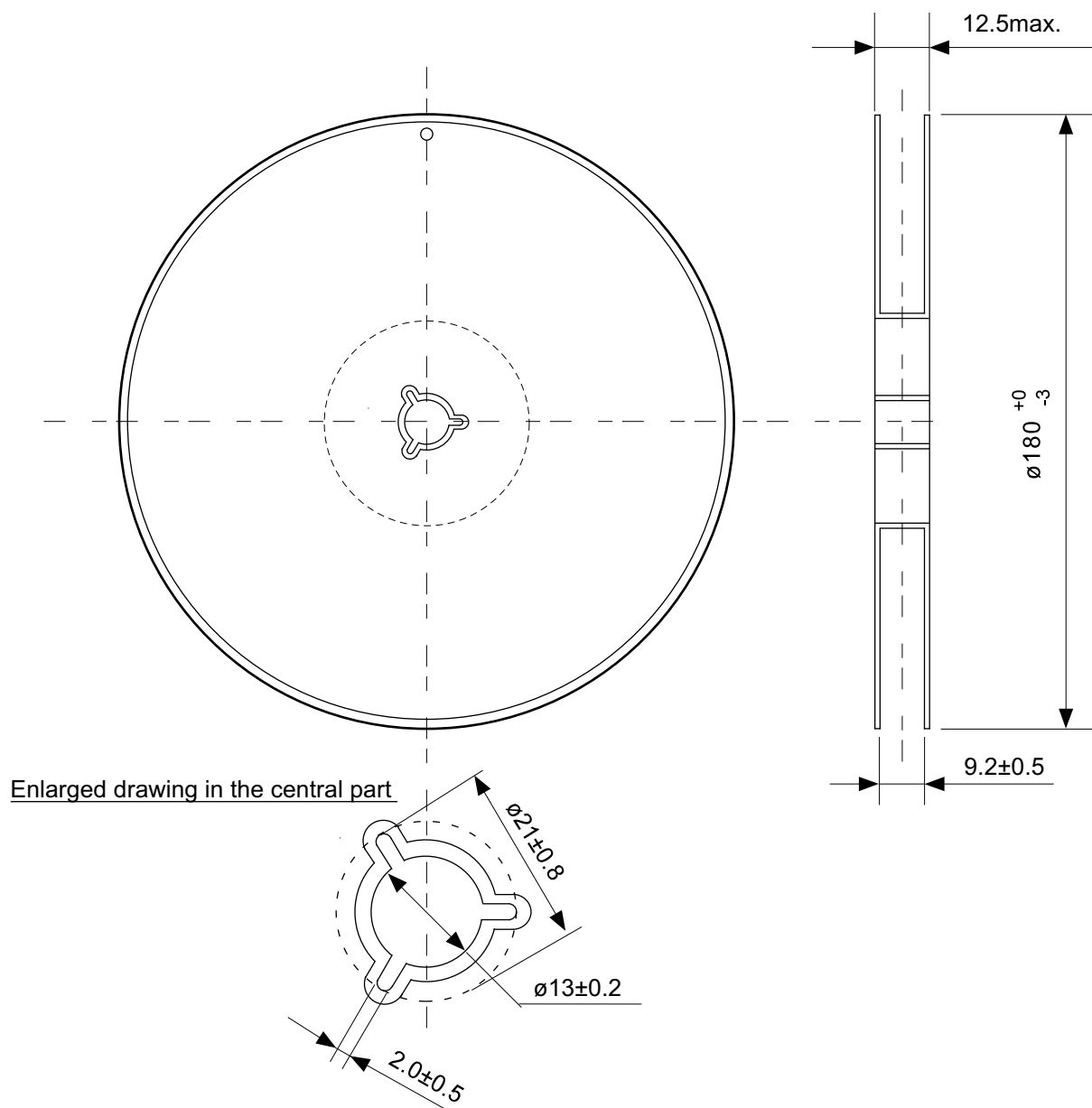
4 3 2 1  
5 6 7 8



Feed direction

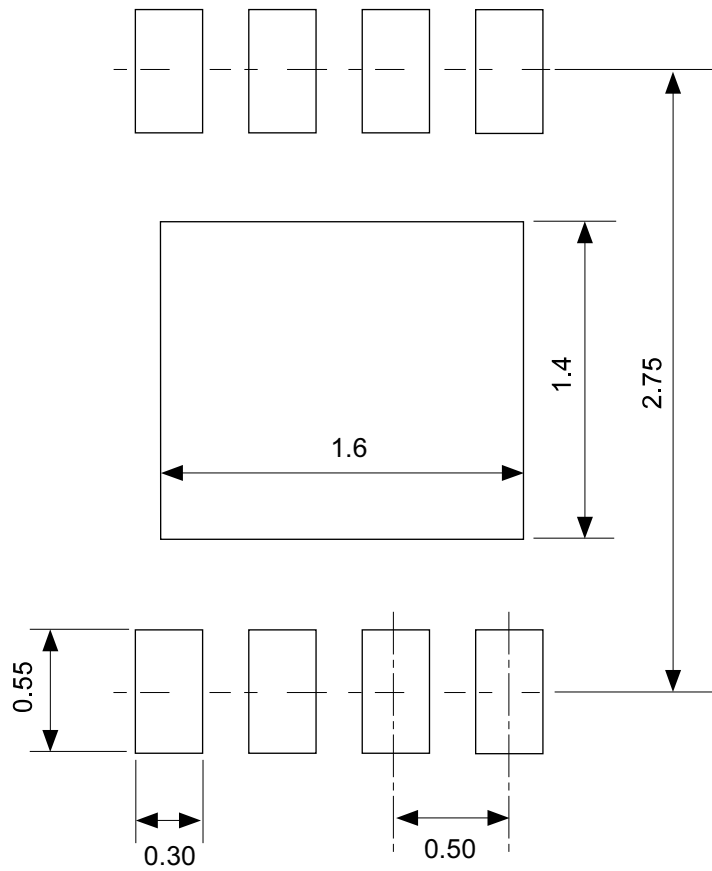
No. PQ008-B-C-SD-1.0

|            |                      |
|------------|----------------------|
| TITLE      | DFN-8-B-Carrier Tape |
| No.        | PQ008-B-C-SD-1.0     |
| ANGLE      |                      |
| UNIT       | mm                   |
|            |                      |
| ABLIC Inc. |                      |



No. PQ008-B-R-SD-1.0

|            |                  |      |       |
|------------|------------------|------|-------|
| TITLE      | DFN-8-B-Reel     |      |       |
| No.        | PQ008-B-R-SD-1.0 |      |       |
| ANGLE      |                  | QTY. | 4,000 |
| UNIT       | mm               |      |       |
|            |                  |      |       |
| ABLIC Inc. |                  |      |       |



No. PQ008-B-L-SD-1.0

|            |                             |
|------------|-----------------------------|
| TITLE      | DFN-8-B-Land Recommendation |
| No.        | PQ008-B-L-SD-1.0            |
| ANGLE      |                             |
| UNIT       | mm                          |
|            |                             |
| ABLIC Inc. |                             |

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